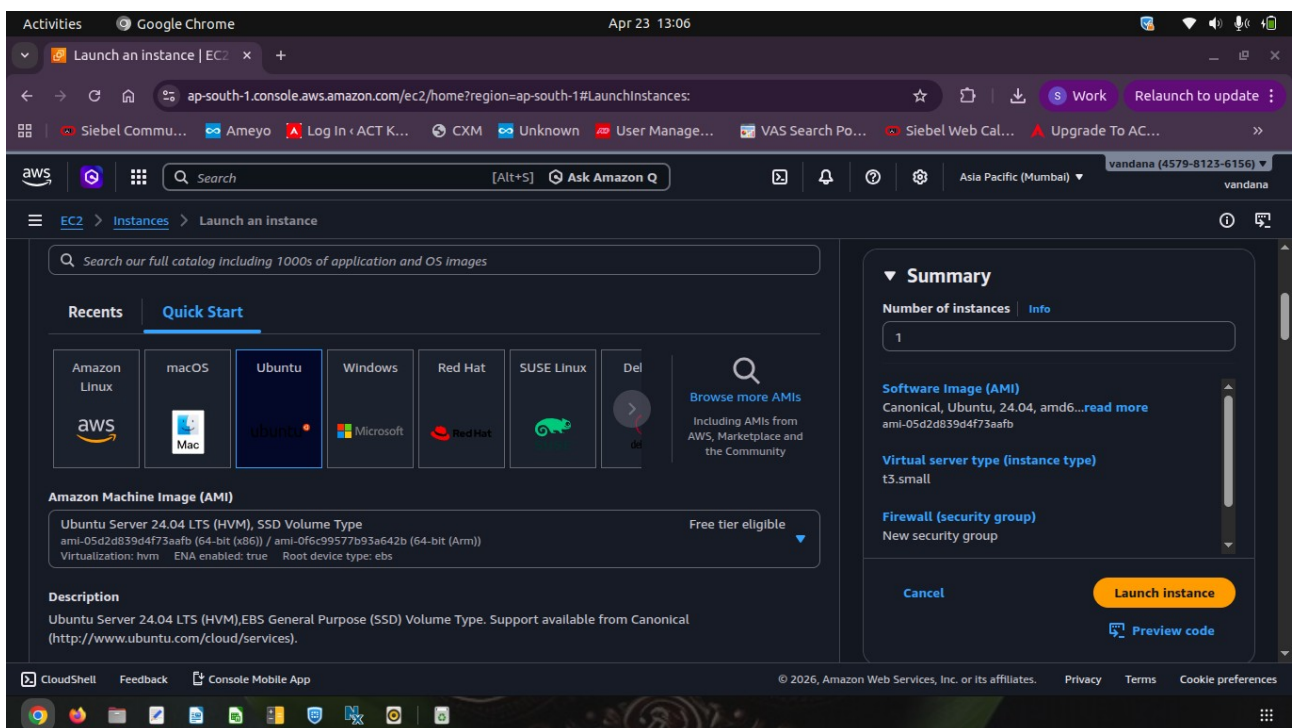
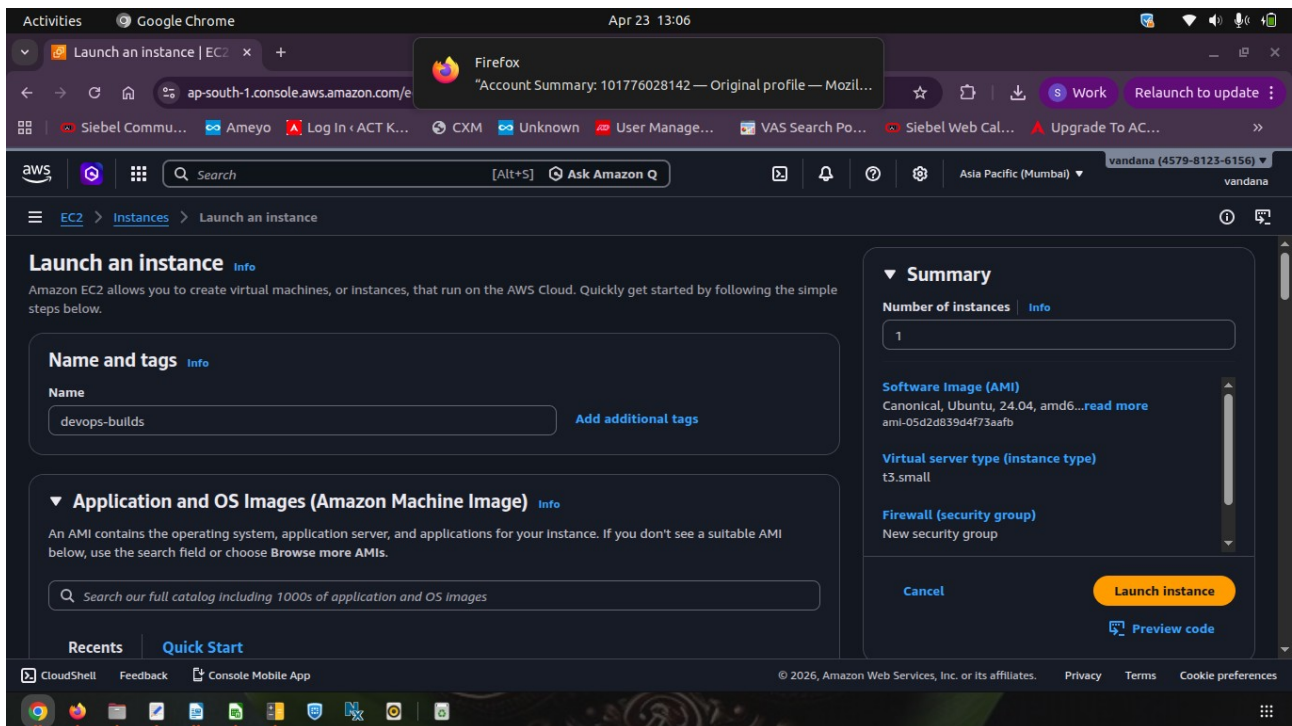
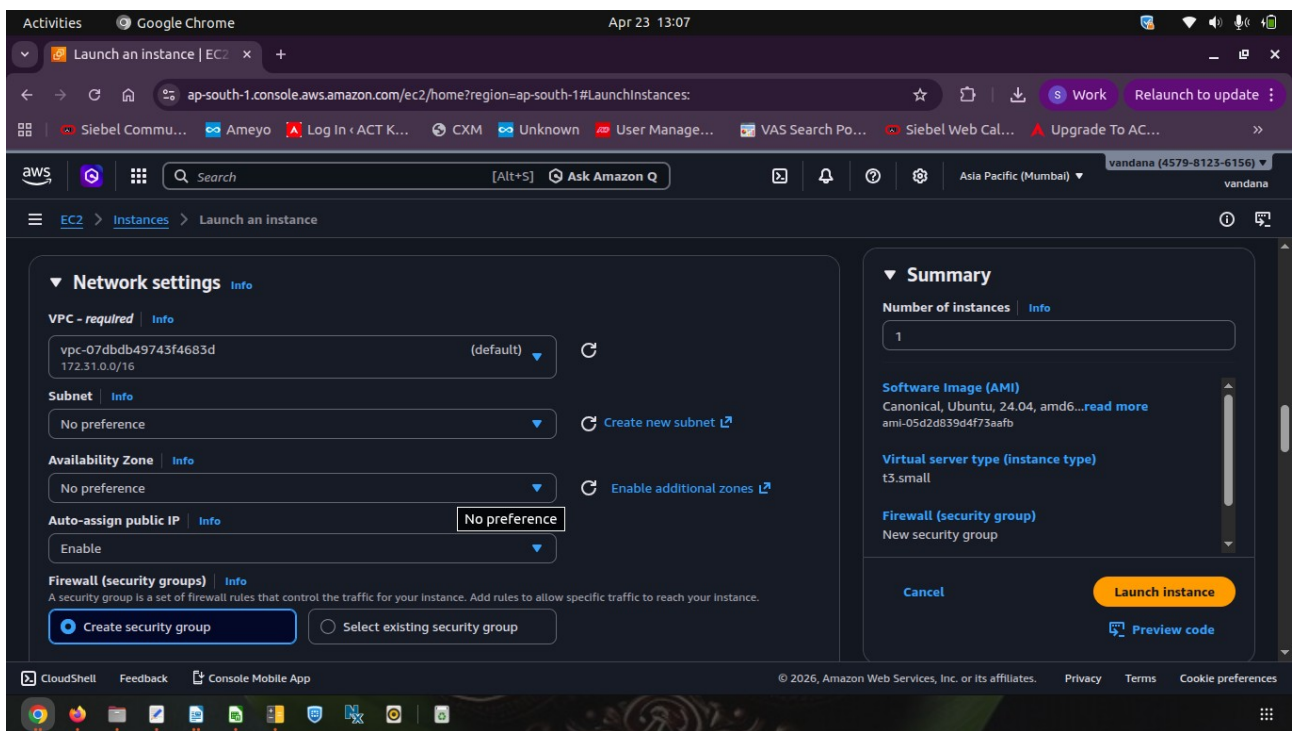
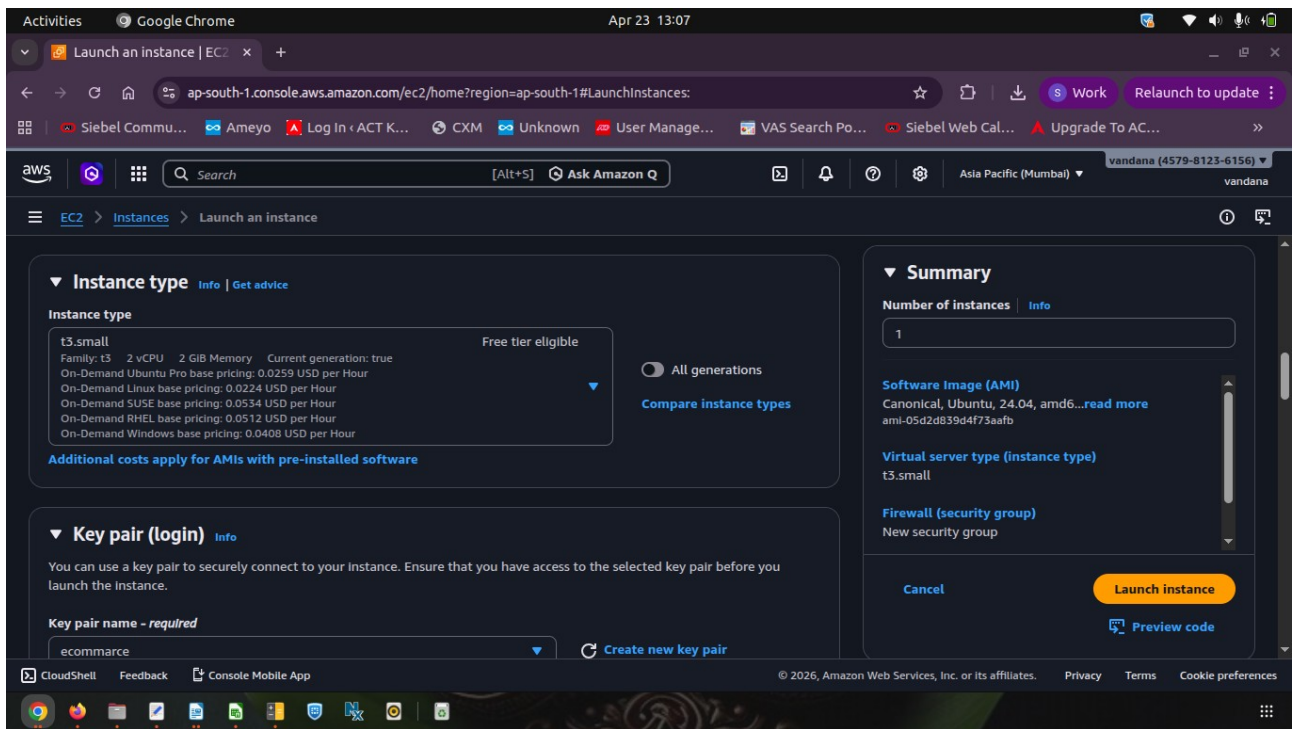
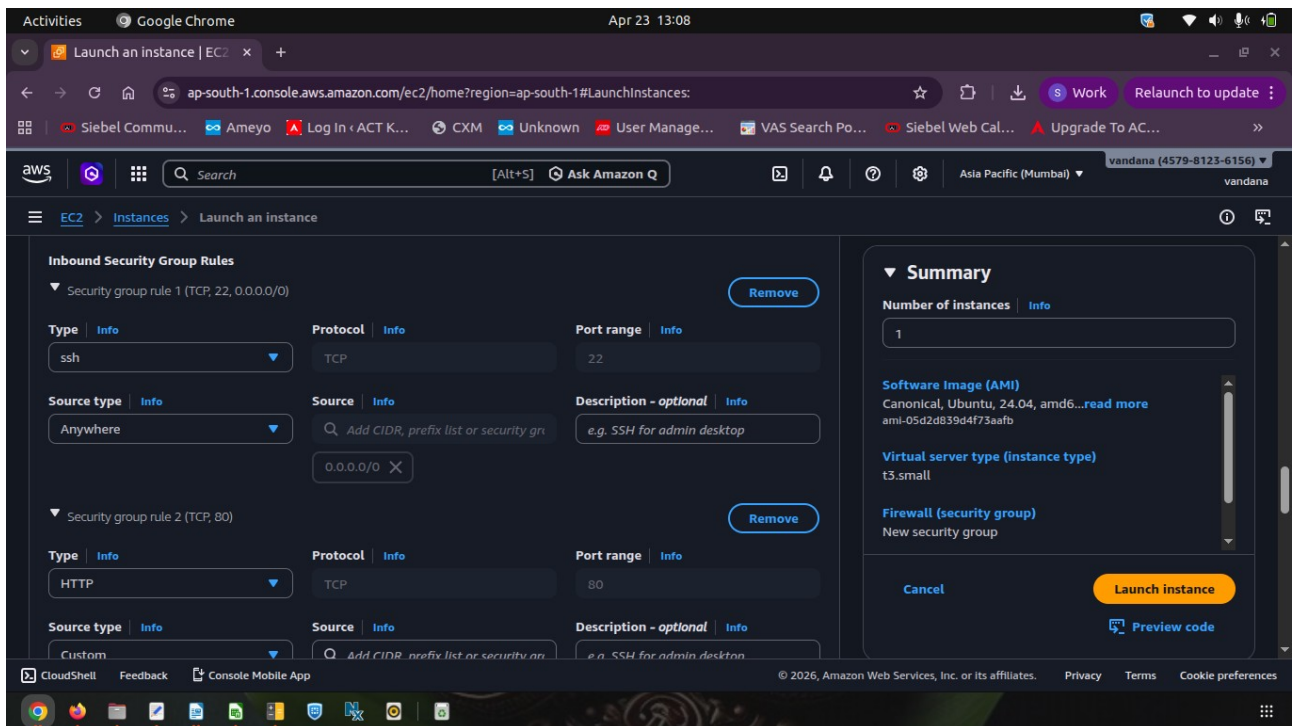




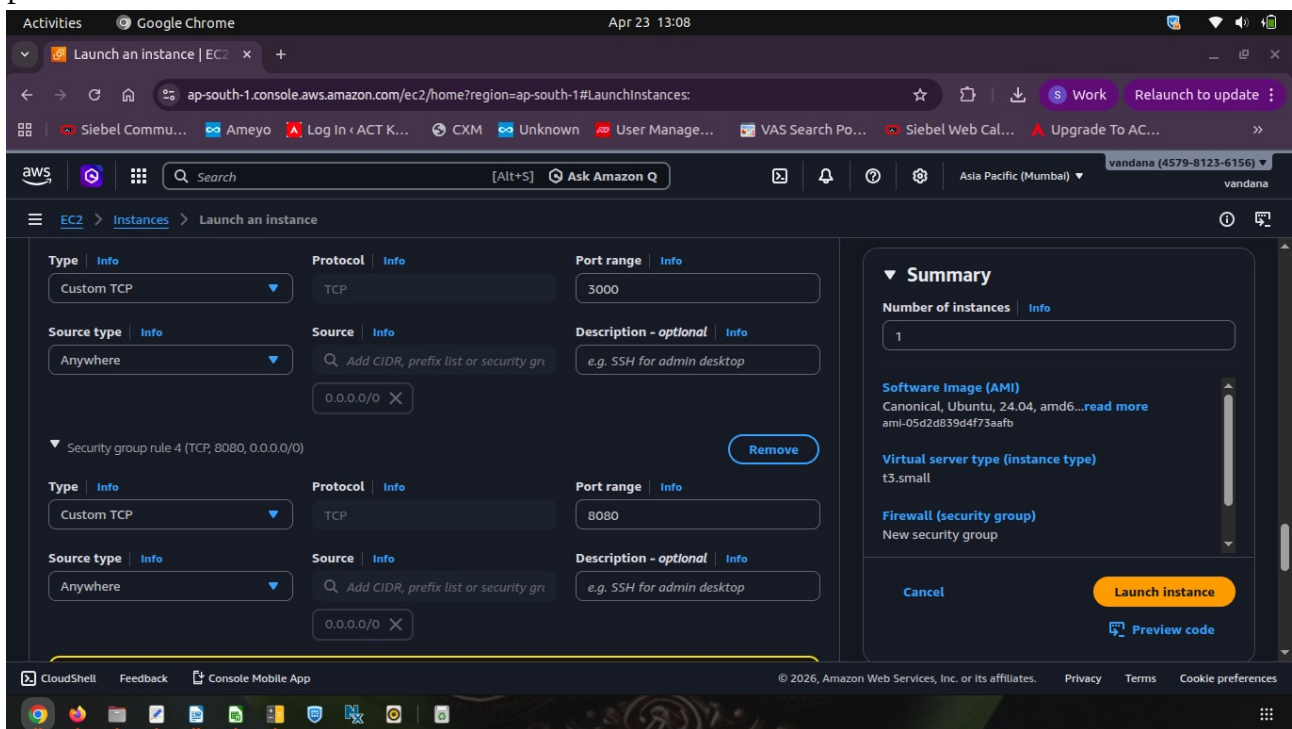
F



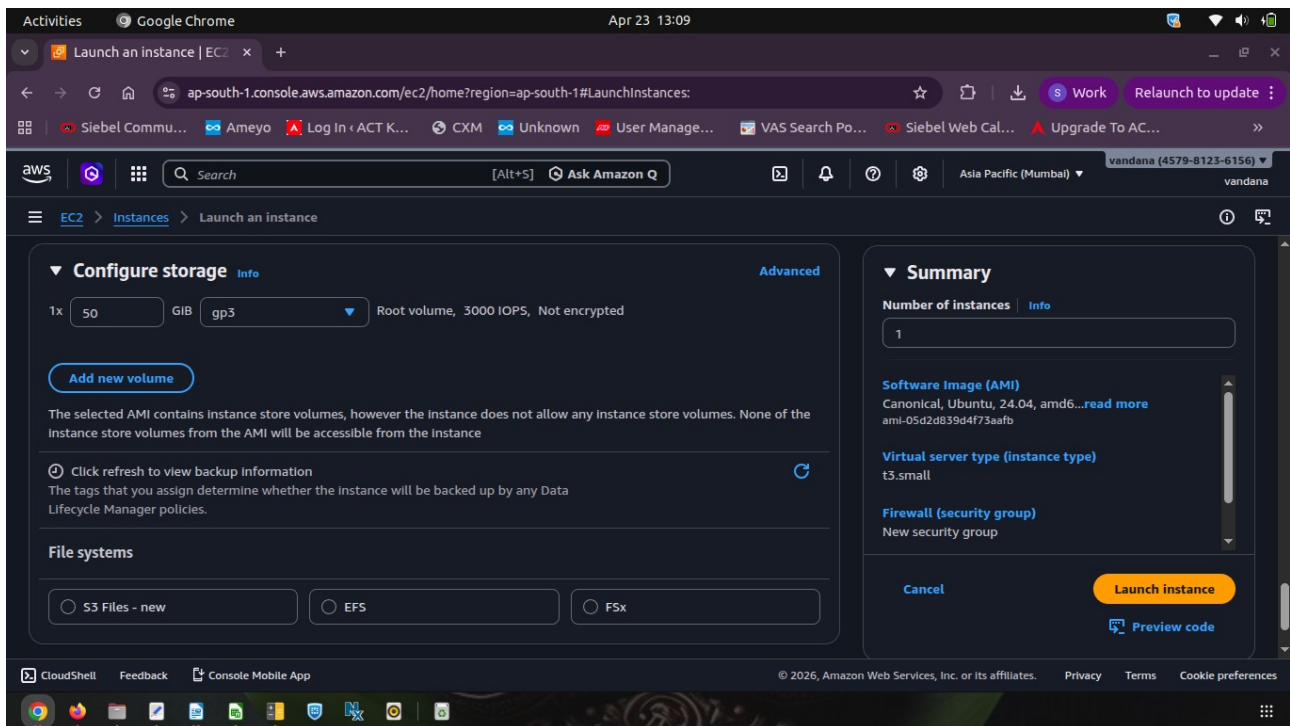
f



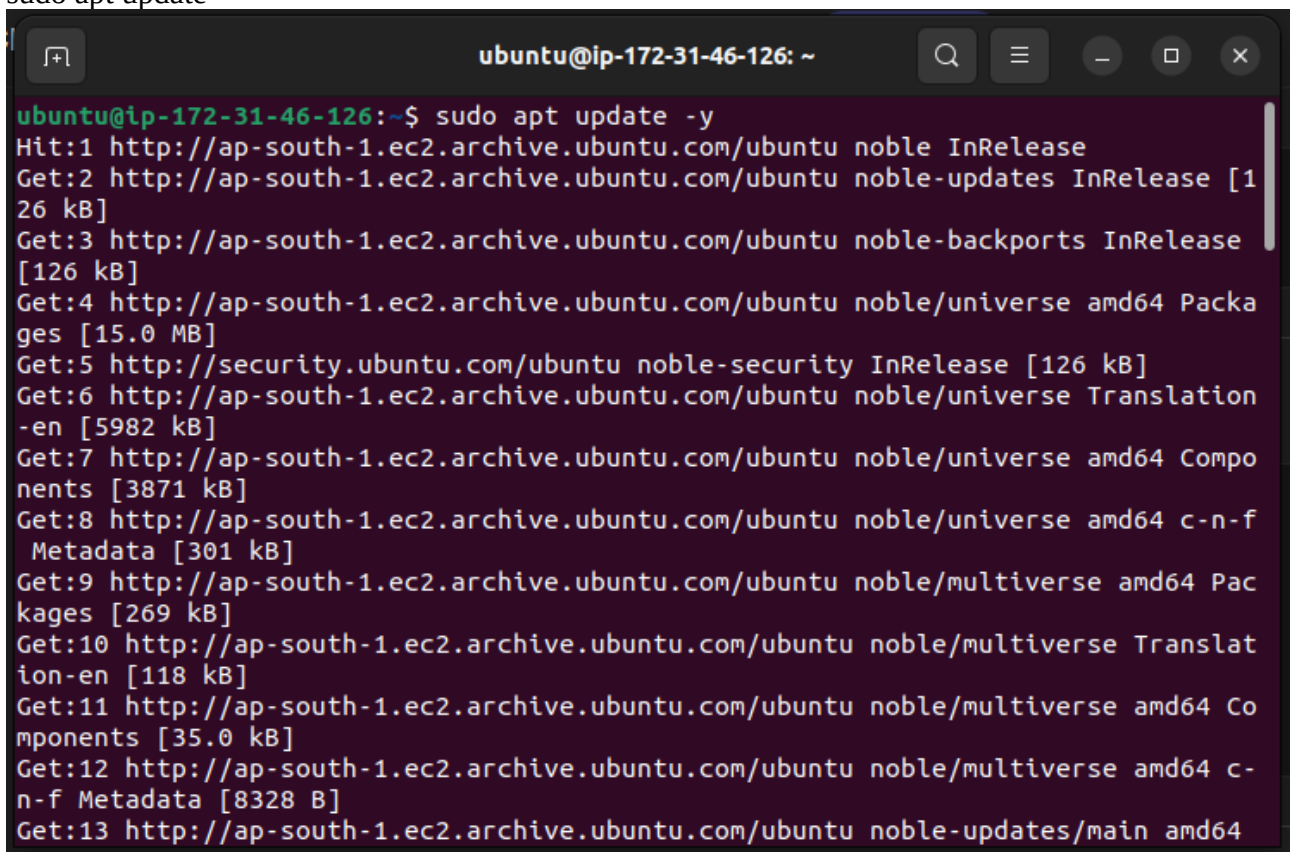
f



f



S
sudo apt update



S

```
ubuntu@ip-172-31-46-126: ~  
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Component  
s [74.2 kB]  
Get:49 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Met  
adata [23.1 kB]  
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Package  
s [2833 kB]  
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-e  
n [666 kB]  
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Compone  
nts [212 B]  
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Package  
s [28.8 kB]  
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-e  
n [7172 B]  
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Compone  
nts [208 B]  
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f M  
etadata [396 B]  
Fetched 41.7 MB in 7s (6349 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
66 packages can be upgraded. Run 'apt list --upgradable' to see them.  
ubuntu@ip-172-31-46-126:~$
```

sudo apt install git

```
ubuntu@ip-172-31-46-126:~$ sudo apt install git -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
git is already the newest version (1:2.43.0-1ubuntu7.3).  
git set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 66 not upgraded.  
ubuntu@ip-172-31-46-126:~$
```

sudo apt install docker.io

```
ubuntu@ip-172-31-46-126: ~  
ubuntu@ip-172-31-46-126:~$ sudo apt install docker.io  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan  
Suggested packages:  
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx  
  docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils  
The following NEW packages will be installed:  
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc  
  ubuntu-fan  
0 upgraded, 8 newly installed, 0 to remove and 66 not upgraded.  
Need to get 73.8 MB of archives.  
After this operation, 279 MB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz  
amd64 2.8-1 [65.6 kB]  
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 bridge-ut  
ils amd64 1.7.1-1ubuntu2 [33.9 kB]  
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 r  
unc amd64 1.3.4-0ubuntu1~24.04.1 [9574 kB]  
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c  
ontainerd amd64 2.2.1-0ubuntu1~24.04.2 [28.1 MB]
```

d

```
ubuntu@ip-172-31-46-126: ~  
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service →  
/usr/lib/systemd/system/ubuntu-fan.service.  
Setting up docker.io (29.1.3-0ubuntu3~24.04.1) ...  
info: Selecting GID from range 100 to 999 ...  
info: Adding group `docker' (GID 113) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /us  
r/lib/systemd/system/docker.service.  
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/li  
b/systemd/system/docker.socket.  
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-46-126:~$
```

d

sudo apt install docker-compose

```
ubuntu@ip-172-31-46-126: ~  
ubuntu@ip-172-31-46-126:~$ sudo apt install docker-compose  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  python3-compose python3-docker python3-dockerpty python3-docopt  
  python3-dotenv python3-texttable python3-websocket  
The following NEW packages will be installed:  
  docker-compose python3-compose python3-docker python3-dockerpty  
  python3-docopt python3-dotenv python3-texttable python3-websocket  
0 upgraded, 8 newly installed, 0 to remove and 66 not upgraded.  
Need to get 297 kB of archives.  
After this operation, 1589 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-websocket all 1.7.0-1 [38.1 kB]  
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-docker all 5.0.3-1ubuntu1.1 [89.1 kB]  
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-dockerpty all 0.4.1-5 [11.4 kB]  
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-docopt all 0.6.2-6 [26.1 kB]  
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-dotenv all 1.0.1-1 [22.3 kB]
```

f

```
ubuntu@ip-172-31-46-126: ~  
Preparing to unpack .../7-docker-compose_1.29.2-6ubuntu1_all.deb ...  
Unpacking docker-compose (1.29.2-6ubuntu1) ...  
Setting up python3-dotenv (1.0.1-1) ...  
Setting up python3-texttable (1.6.7-1) ...  
Setting up python3-docopt (0.6.2-6) ...  
Setting up python3-websocket (1.7.0-1) ...  
Setting up python3-dockerpty (0.4.1-5) ...  
Setting up python3-docker (5.0.3-1ubuntu1.1) ...  
Setting up python3-compose (1.29.2-6ubuntu1) ...  
Setting up docker-compose (1.29.2-6ubuntu1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-46-126:~$
```

f


```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~$ sudo systemctl start docker
ubuntu@ip-172-31-46-126:~$ sudo systemctl enable docker
ubuntu@ip-172-31-46-126:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-46-126:~$ newgrp docker
ubuntu@ip-172-31-46-126:~$ git clone https://github.com/sriram-R-krishnan/devops-build.git
Cloning into 'devops-build'...
remote: Enumerating objects: 21, done.
remote: Total 21 (delta 0), reused 0 (delta 0), pack-reused 21 (from 1)
Receiving objects: 100% (21/21), 720.09 KiB | 18.95 MiB/s, done.
ubuntu@ip-172-31-46-126:~$ cd devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ npm install
Command 'npm' not found, but can be installed with:
sudo apt install npm
ubuntu@ip-172-31-46-126:~/devops-build$ nano Dockerfile
ubuntu@ip-172-31-46-126:~/devops-build$ cat Dockerfile
FROM node:18

WORKDIR /app

COPY package*.json ./
RUN npm install

COPY . .
```

c
Create dockerfile

```
ubuntu@ip-172-31-46-126: ~/devops-build
Receiving objects: 100% (21/21), 720.09 KiB | 18.95 MiB/s, done.
ubuntu@ip-172-31-46-126:~$ cd devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ npm install
Command 'npm' not found, but can be installed with:
sudo apt install npm
ubuntu@ip-172-31-46-126:~/devops-build$ nano Dockerfile
ubuntu@ip-172-31-46-126:~/devops-build$ cat Dockerfile
FROM node:18

WORKDIR /app

COPY package*.json ./
RUN npm install

COPY . .

RUN npm run build

RUN npm install -g serve

EXPOSE 80

CMD ["serve", "-s", "build", "-l", "80"]
ubuntu@ip-172-31-46-126:~/devops-build$
```

d


```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ nano Dockerfile
ubuntu@ip-172-31-46-126:~/devops-build$ cat Dockerfile
FROM nginx:alpine

# Remove default nginx static files
RUN rm -rf /usr/share/nginx/html/*

# Copy your build files
COPY build/ /usr/share/nginx/html/

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
ubuntu@ip-172-31-46-126:~/devops-build$ cat docker-compose.yml
version: "3"

services:
  react-app:
    build: .
    ports:
      - "80:80"
ubuntu@ip-172-31-46-126:~/devops-build$
```

sudo apt install npm

```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ sudo apt install npm
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential bzip2 cpp
  cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev eslint fakeroot
  fontconfig-config fonts-dejavu-core fonts-dejavu-mono g++ g++-13
  g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base
  gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu gyp handlebars
  javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libaom3 libasan8 libatomic1 libauthen-sasl-perl
  libbinutils libc-dev-bin libc-devtools libc6-dev libcares2 libcc1-0
  libclone-perl libcrypt-dev libctf-nobfd0 libctf0 libdata-dump-perl
  libde265-0 libdeflate0 libdpkg-perl libdrm-amdgpu1 libdrm-intel1
  libegl-mesa0 libegl1 libencode-locale-perl libfakeroot libfile-basedir-perl
  libfile-desktopentry-perl libfile-fcntllock-perl libfile-listing-perl
  libfile-mimeinfo-perl libfont-afm-perl libfontconfig1 libgbm1 libgcc-13-dev
  libgd3 libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common
  libgl1 libgl1-mesa-dri libgles2 libglvnd0 libglx-mesa0 libglx0 libgomp1
  libgprofng0 libheif-plugin-aomdec libheif-plugin-aomenc
  libheif-plugin-libde265 libheif1 libhtml-form-perl libhtml-format-perl
  libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl
  libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl
```

d

```
ubuntu@ip-172-31-46-126: ~/devops-build
node-ajv-keywords node-ampproject-remapping node-ansi-escapes
node-ansi-regex node-ansi-styles node-anymatch node-aproba node-archy
node-are-we-there-yet node-argparse node-arrify node-assert node-async
node-async-each node-auto-bind node-babel-helper-define-polyfill-provider
node-babel-plugin-add-module-exports node-babel-plugin-lodash
node-babel-plugin-polyfill-corejs2 node-babel-plugin-polyfill-corejs3
node-babel-plugin-polyfill-regenerator node-babel7 node-babel7-runtime
node-balanced-match node-base node-base64-js node-binary-extensions
node-brace-expansion node-braces node-browserslist node-builtins node-busboy
node-cacache node-cache-base node-camelcase node-caniuse-lite node-chalk
node-chokidar node-chownr node-chrome-trace-event node-ci-info
node-cjs-module-lexer node-cli-boxes node-cli-cursor node-cli-table
node-cli-truncate node-cliui node-clone node-clone-deep
node-collection-visit node-color-convert node-color-name node-colors
node-columnify node-commander node-commandir node-concat-stream
node-console-control-strings node-convert-source-map node-copy-concurrently
node-core-js node-core-js-compat node-core-js-pure node-core-util-is
node-coveralls node-css-loader node-css-selector-tokenizer
node-data-uri-to-buffer node-debbundle-es-to-primitive node-debug
node-decamelize node-decompress-response node-deep-equal node-deep-is
node-defaults node-define-properties node-define-property node-defined
node-del node-delegates node-depd node-diff node-doctrine
node-electron-to-chromium node-encoding node-end-of-stream
node-enhanced-resolve node-err-code node-errno node-error-ex
```

d

```
ubuntu@ip-172-31-46-126: ~/devops-build
x11-utils x11-xserver-utils xdg-utils zutty
Suggested packages:
binutils-doc gprofng-gui bzip2-doc cpp-doc gcc-13-locales cpp-13-doc
debian-keyring node-babel-eslint node-esprima-fb node-inquirer g++-multilib
g++-13-multilib gcc-13-doc gcc-multilib autoconf automake libtool flex bison
gdb gcc-doc gcc-13-multilib gdb-x86-64-linux-gnu apache2 | lighttpd | httpd
libdigest-hmac-perl libgssapi-perl glibc-doc bzip libgd-tools
libheif-plugin-x265 libheif-plugin-ffmpegdec libheif-plugin-jpegdec
libheif-plugin-jpegenc libheif-plugin-j2kdec libheif-plugin-j2kenc
libheif-plugin-rav1e libheif-plugin-svtenc libio-compress-brotli-perl
libjs-angularjs libcrypt-ssleay-perl notification-daemon libssl-doc
libstdc++-13-doc libsub-name-perl libbusiness-isbn-perl libregexp-ipv6-perl
libauthen-ntlm-perl libunicode-map8-perl libunicode-string-perl
xml-twig-tools make-doc node-babel-plugin-polyfill-es-shims
node-babel7-debug livescript chai node-jest-diff debhelper mesa-utils nickle
cairo-5c xorg-docs-core
Recommended packages:
luit
The following NEW packages will be installed:
binutils binutils-common binutils-x86-64-linux-gnu build-essential bzip2 cpp
cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev eslint fakeroot
fontconfig-config fonts-dejavu-core fonts-dejavu-mono g++ g++-13
g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base
gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu gyp handlebars
```

d

```
ubuntu@ip-172-31-46-126: ~/devops-build
node-are-we-there-yet node-argparse node-arrify node-assert node-async
node-async-each node-auto-bind node-babel-helper-define-polyfill-provider
node-babel-plugin-add-module-exports node-babel-plugin-lodash
node-babel-plugin-polyfill-corejs2 node-babel-plugin-polyfill-corejs3
node-babel-plugin-polyfill-regenerator node-babel7 node-babel7-runtime
node-balanced-match node-base node-base64-js node-binary-extensions
node-brace-expansion node-braces node-browserslist node-builtins node-busboy
node-cacache node-cache-base node-camelcase node-caniuse-lite node-chalk
node-chokidar node-chownr node-chrome-trace-event node-ci-info
node-cjs-module-lexer node-cli-boxes node-cli-cursor node-cli-table
node-cli-truncate node-clui node-clone node-clone-deep
node-collection-visit node-color-convert node-color-name node-colors
node-columnify node-commander node-commander node-concat-stream
node-console-control-strings node-convert-source-map node-copy-concurrently
node-core-js node-core-js-compat node-core-js-pure node-core-util-is
node-coveralls node-css-loader node-css-selector-tokenizer
node-data-uri-to-buffer node-debbundle-es-to-primitive node-debug
node-decamelize node-decompress-response node-deep-equal node-deep-is
node-defaults node-define-properties node-define-property node-defined
node-del node-delegates node-depd node-diff node-doctrine
node-electron-to-chromium node-encoding node-end-of-stream
node-enhanced-resolve node-err-code node-errno node-error-ex
node-es-abstract node-es-module-lexer node-es6-error
node-escape-string-regexp node-escapegen node-eslint-scope node-eslint-utils
```

d

```
ubuntu@ip-172-31-46-126: ~/devops-build
libssl3t64 amd64 3.0.13-0ubuntu3.9 [1941 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 o
penssl amd64 3.0.13-0ubuntu3.9 [1002 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 b
inutils-common amd64 2.42-4ubuntu2.10 [240 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 l
ibsqlite3 amd64 3.45.1-1ubuntu0.1 [817 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 l
ibbinutils amd64 2.42-4ubuntu2.10 [577 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 l
ibctf-nobfd0 amd64 2.42-4ubuntu2.10 [98.0 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 l
ibctf0 amd64 2.42-4ubuntu2.10 [94.5 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 l
ibgprofng0 amd64 2.42-4ubuntu2.10 [849 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 b
inutils-x86-64-linux-gnu amd64 2.42-4ubuntu2.10 [2463 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
binutils amd64 2.42-4ubuntu2.10 [18.2 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
libc-dev-bin amd64 2.39-0ubuntu8.7 [20.4 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
linux-libc-dev amd64 6.8.0-110.110 [1587 kB]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libcrypt
```

d


```
ubuntu@ip-172-31-46-126: ~/devops-build
all 9.2.0~ds1-2 [631 kB]
Fetched 176 MB in 3s (57.4 MB/s)
Extracting templates from packages: 100%
(Reading database ... 73199 files and directories currently installed.)
Preparing to unpack .../libssl3t64_3.0.13-0ubuntu3.9_amd64.deb ...
Unpacking libssl3t64:amd64 (3.0.13-0ubuntu3.9) over (3.0.13-0ubuntu3.7) ...
Setting up libssl3t64:amd64 (3.0.13-0ubuntu3.9) ...
(Reading database ... 73199 files and directories currently installed.)
Preparing to unpack .../000-openssl_3.0.13-0ubuntu3.9_amd64.deb ...
Unpacking openssl (3.0.13-0ubuntu3.9) over (3.0.13-0ubuntu3.7) ...
Selecting previously unselected package binutils-common:amd64.
Preparing to unpack .../001-binutils-common_2.42-4ubuntu2.10_amd64.deb ...
Unpacking binutils-common:amd64 (2.42-4ubuntu2.10) ...
Selecting previously unselected package libsframe1:amd64.
Preparing to unpack .../002-libsframe1_2.42-4ubuntu2.10_amd64.deb ...
Unpacking libsframe1:amd64 (2.42-4ubuntu2.10) ...
Selecting previously unselected package libbinutils:amd64.
Preparing to unpack .../003-libbinutils_2.42-4ubuntu2.10_amd64.deb ...
Unpacking libbinutils:amd64 (2.42-4ubuntu2.10) ...
Selecting previously unselected package libctf-nobfd0:amd64.
Preparing to unpack .../004-libctf-nobfd0_2.42-4ubuntu2.10_amd64.deb ...
Unpacking libctf-nobfd0:amd64 (2.42-4ubuntu2.10) ...
Selecting previously unselected package libctf0:amd64.
Preparing to unpack .../005-libctf0_2.42-4ubuntu2.10_amd64.deb ...
```

d

```
ubuntu@ip-172-31-46-126: ~/devops-build
Setting up libc-devtools (2.39-0ubuntu8.7) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart ssh.service systemd-journald.service systemd-networkd.service
systemd-resolved.service systemd-udevd.service udisks2.service

Service restarts being deferred:
systemctl restart systemd-logind.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #3: sshd[1081]
ubuntu @ user manager service: systemd[1086]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-46-126:~/devops-build$
```

d

Create .dockerignore & Docker Compose

```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ nano .dockerignore
ubuntu@ip-172-31-46-126:~/devops-build$ cat .dockerignore
node_modules
.git
.gitignore
Dockerfile
docker-compose.yml
ubuntu@ip-172-31-46-126:~/devops-build$ nano docker-compose.yml
ubuntu@ip-172-31-46-126:~/devops-build$ cat docker-compose.yml
version: "3"

services:
  react-app:
    build: .
    ports:
      - "80:80"
ubuntu@ip-172-31-46-126:~/devops-build$
```

d

```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ docker-compose down
docker-compose up -d --build
Removing network devops-build_default
Creating network "devops-build_default" with the default driver
Building react-app
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   2.62MB
Step 1/5 : FROM nginx:alpine
alpine: Pulling from library/nginx
aee4e54b3865: Pulling fs layer
82736a35d0e7: Pulling fs layer
781ff50d2644: Pulling fs layer
453da7dbc73e: Pulling fs layer
583599bb7d38: Pulling fs layer
4a8b0b2a5b19: Pulling fs layer
612c0c1df4c5: Pulling fs layer
6a0ac1617861: Pulling fs layer
6192e1e6a438: Download complete
e8fc446e336c: Download complete
453da7dbc73e: Download complete
```

d

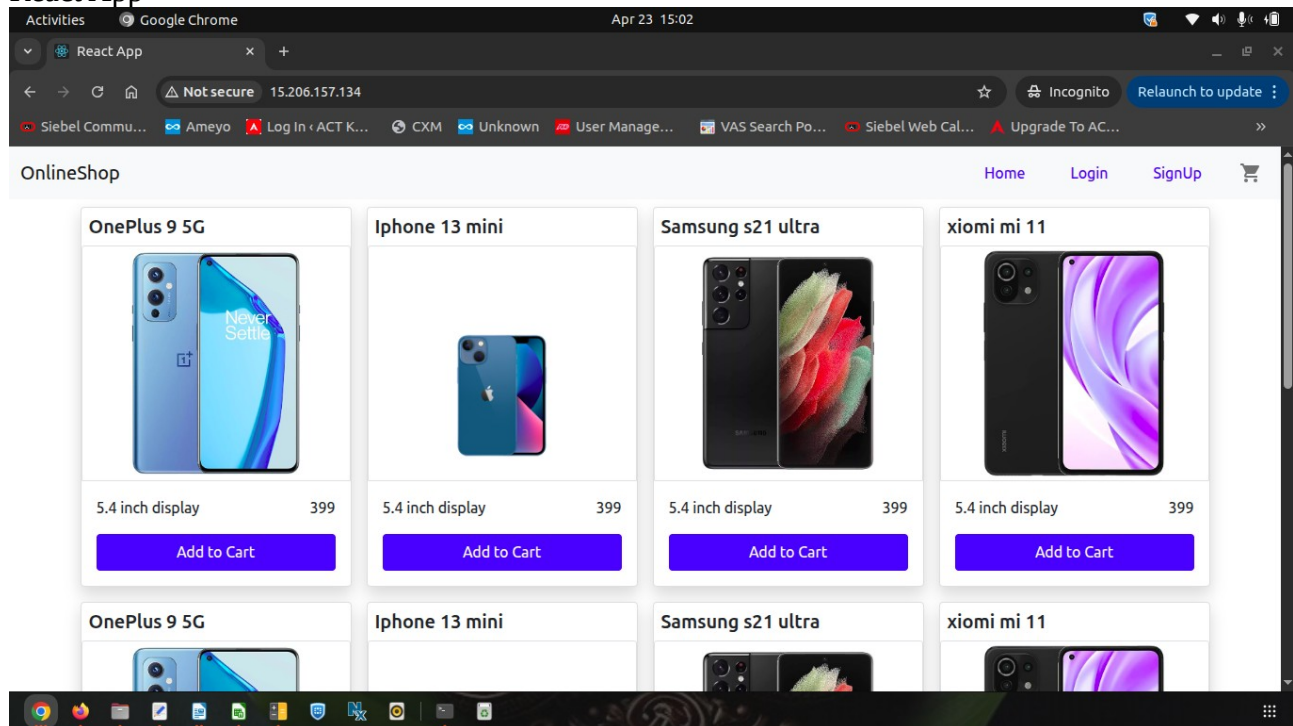
```
ubuntu@ip-172-31-46-126: ~/devops-build
4a8b0b2a5b19: Pull complete
781ff50d2644: Pull complete
612c0c1df4c5: Pull complete
Digest: sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
Status: Downloaded newer image for nginx:alpine
---> 5616878291a2
Step 2/5 : RUN rm -rf /usr/share/nginx/html/*
---> Running in f9e3325545b1
---> Removed intermediate container f9e3325545b1
---> 57d4fd33961b
Step 3/5 : COPY build/ /usr/share/nginx/html/
---> 745cd9091150
Step 4/5 : EXPOSE 80
---> Running in ed348d462ced
---> Removed intermediate container ed348d462ced
---> 68ecedeeed0e
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
---> Running in 39793330fa95
---> Removed intermediate container 39793330fa95
---> a6742b90955f
Successfully built a6742b90955f
Successfully tagged devops-build_react-app:latest
Creating devops-build_react-app_1 ... done
ubuntu@ip-172-31-46-126:~/devops-build$
```

r

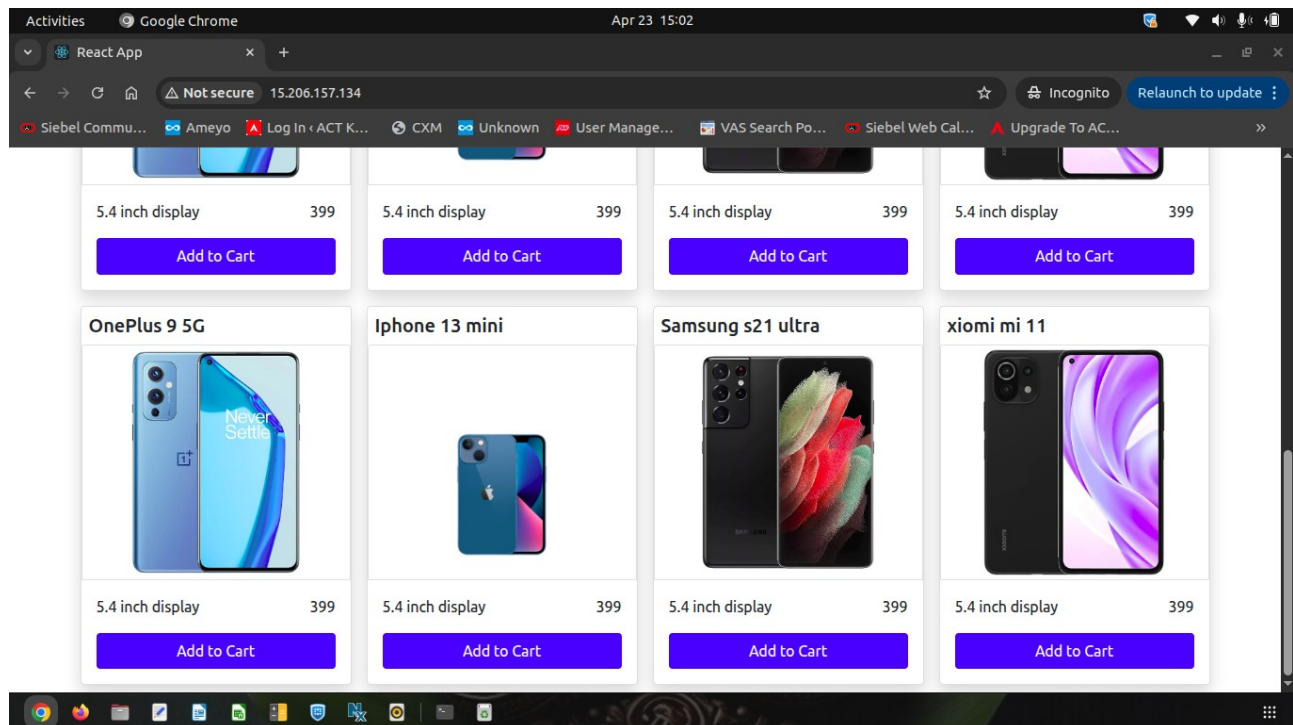
```
ubuntu@ip-172-31-46-126: ~/devops-build
---> 5616878291a2
Step 2/5 : RUN rm -rf /usr/share/nginx/html/*
---> Running in f9e3325545b1
---> Removed intermediate container f9e3325545b1
---> 57d4fd33961b
Step 3/5 : COPY build/ /usr/share/nginx/html/
---> 745cd9091150
Step 4/5 : EXPOSE 80
---> Running in ed348d462ced
---> Removed intermediate container ed348d462ced
---> 68ecedeeed0e
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
---> Running in 39793330fa95
---> Removed intermediate container 39793330fa95
---> a6742b90955f
Successfully built a6742b90955f
Successfully tagged devops-build_react-app:latest
Creating devops-build_react-app_1 ... done
ubuntu@ip-172-31-46-126:~/devops-build$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED
STATUS        PORTS
5f9ab5c4fbaa   devops-build_react-app              "/docker-entrypoint...." 32 minutes ago
Up 32 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp  devops-build_react-app_1
ubuntu@ip-172-31-46-126:~/devops-build$
```

The application repository contained pre-built static assets, so instead of using Node.js, I used Nginx to serve the build folder, which is the recommended production approach.

React App

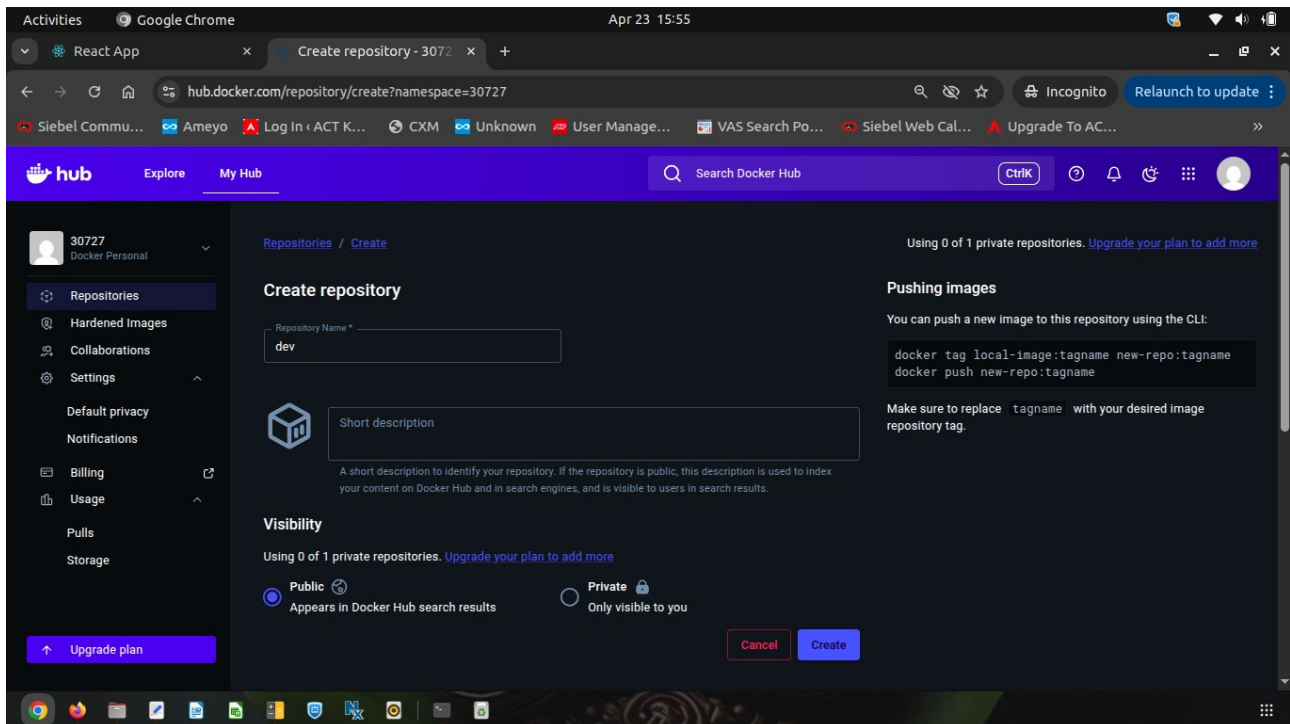


f

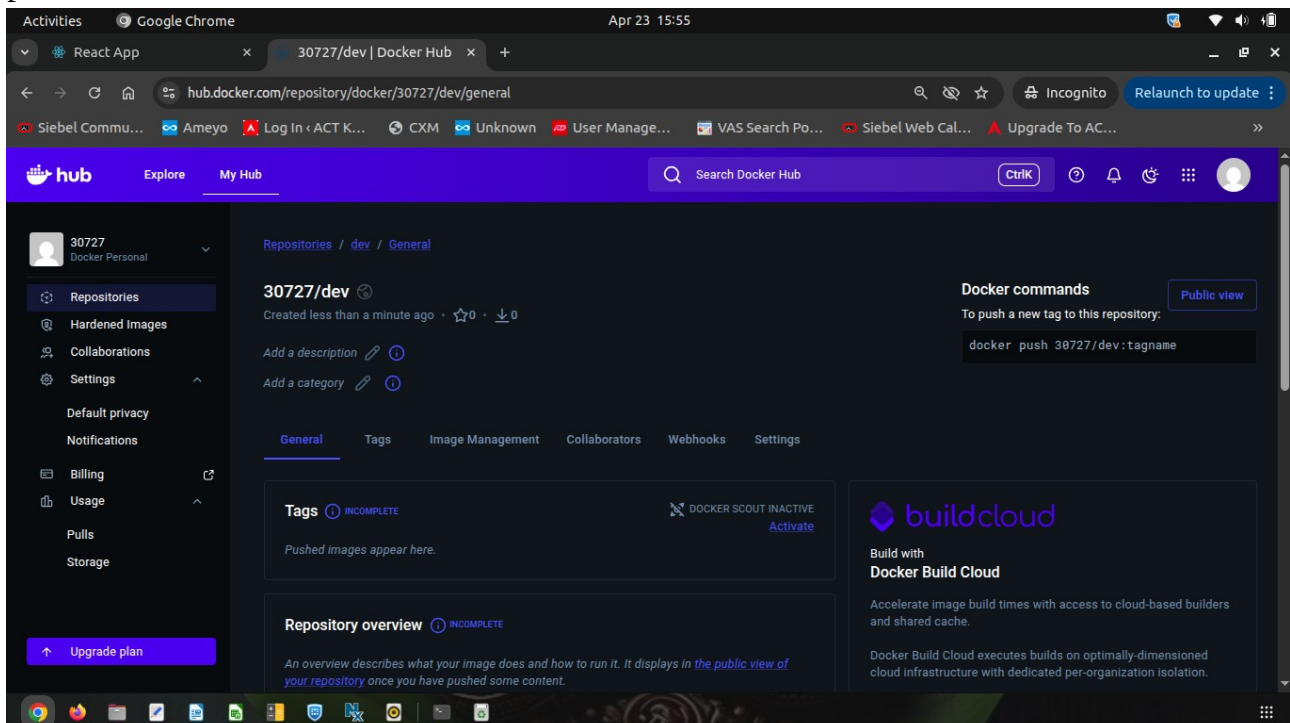


f

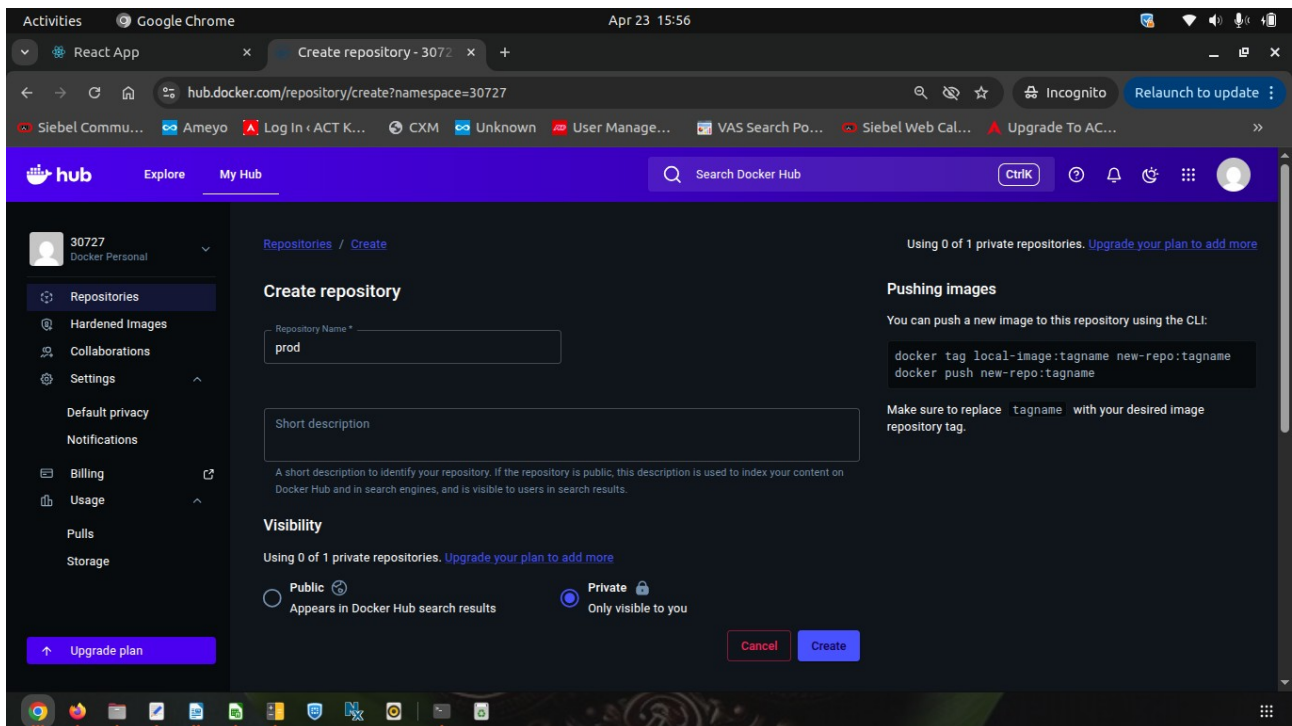
Docker Hub + Scripts
Repositories --
30727/dev → Public
30727/prod → Private



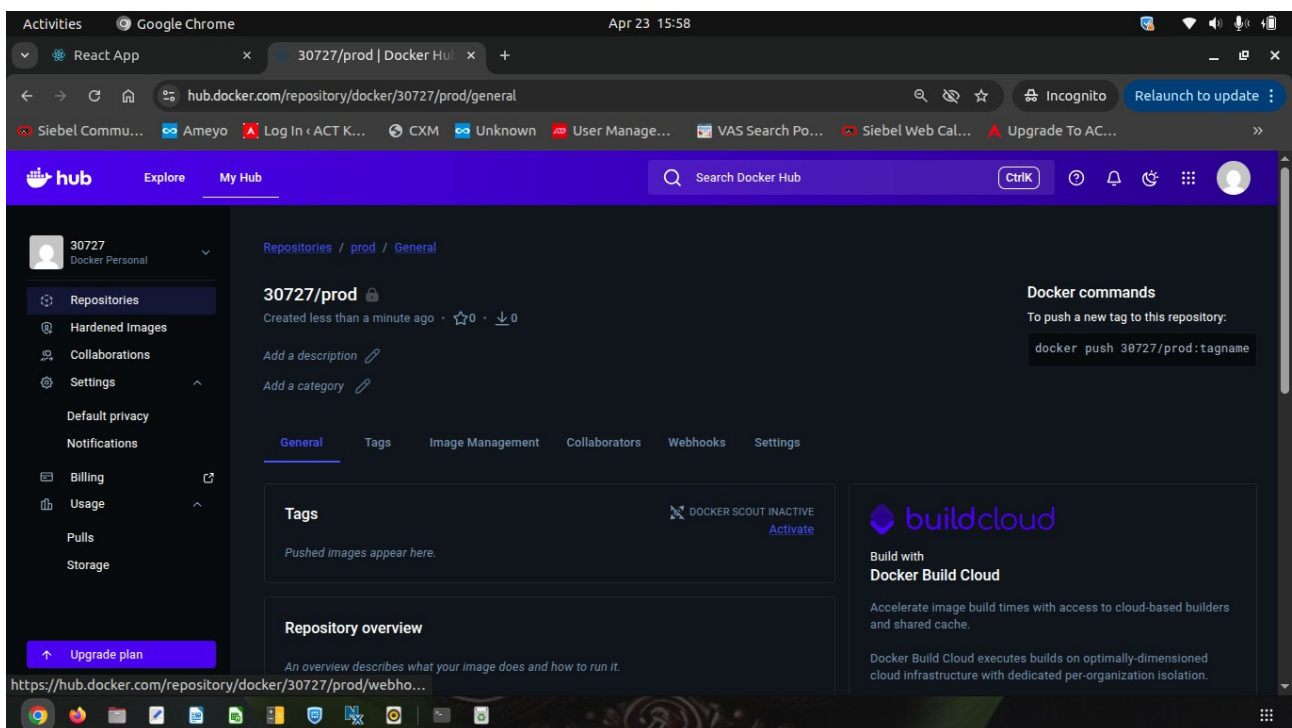
f



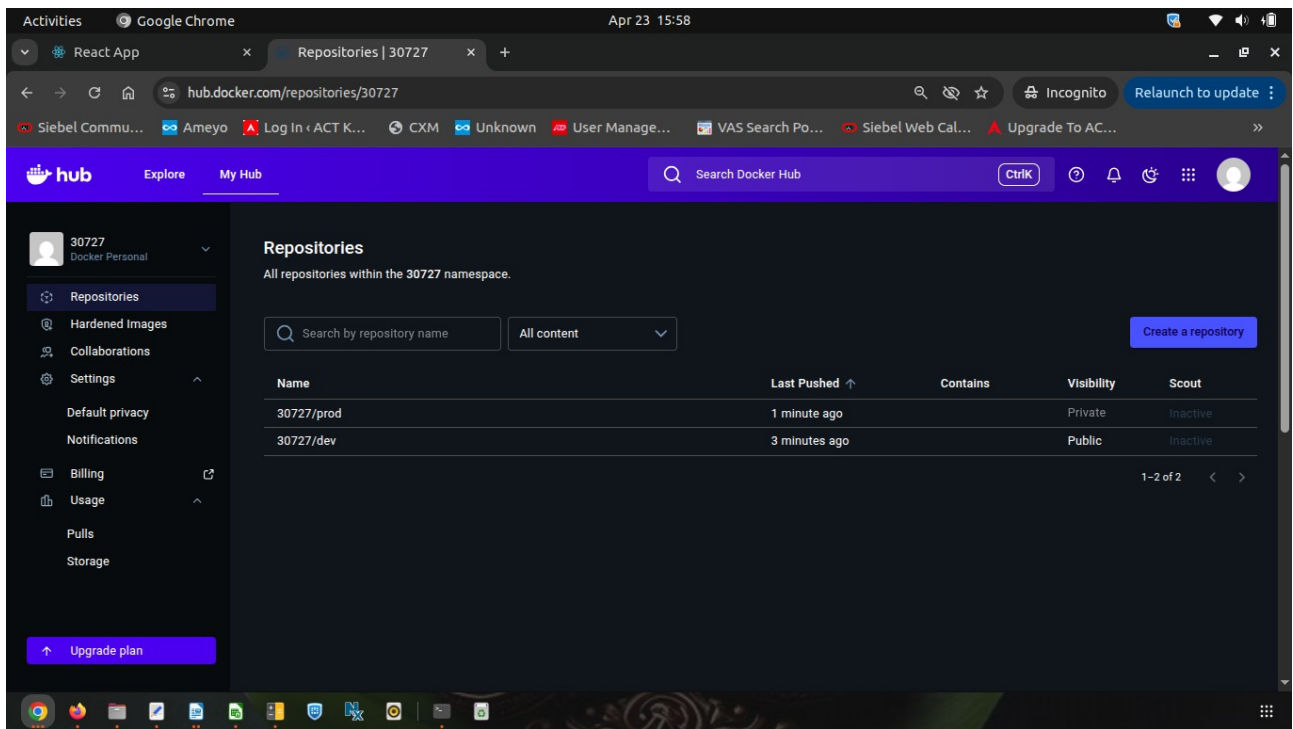
f



f

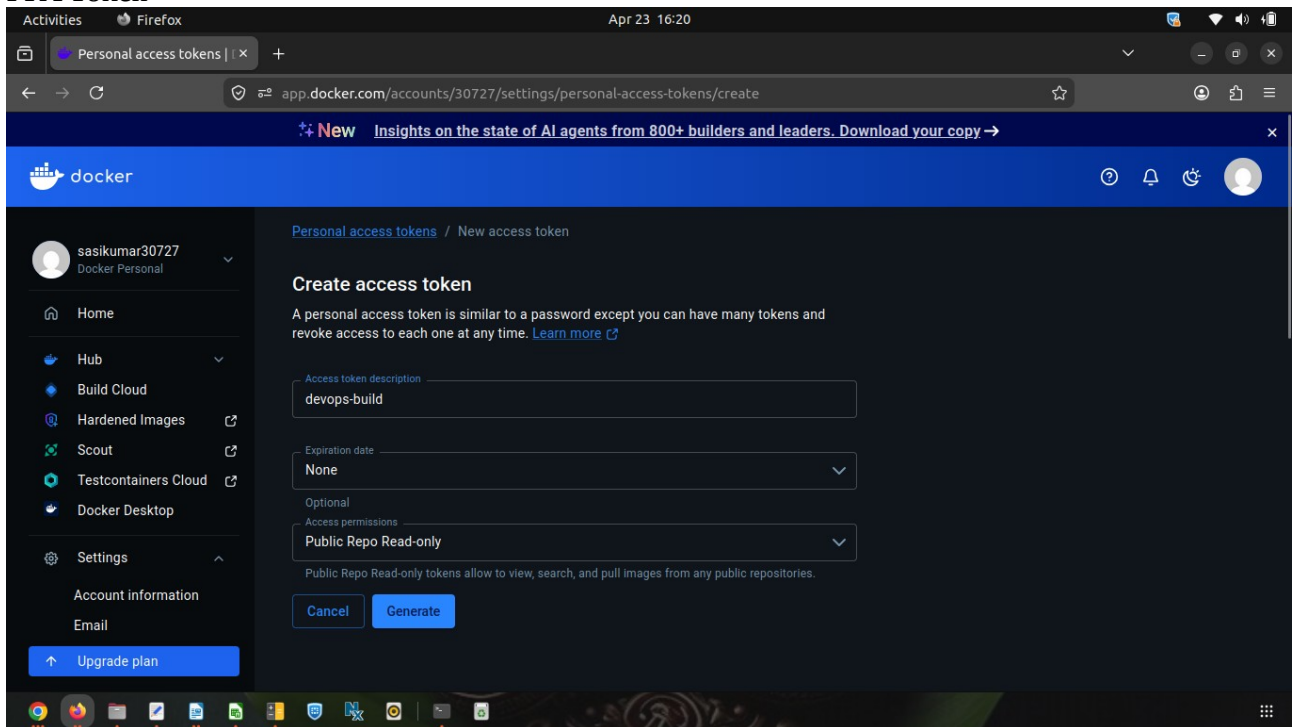


f



f

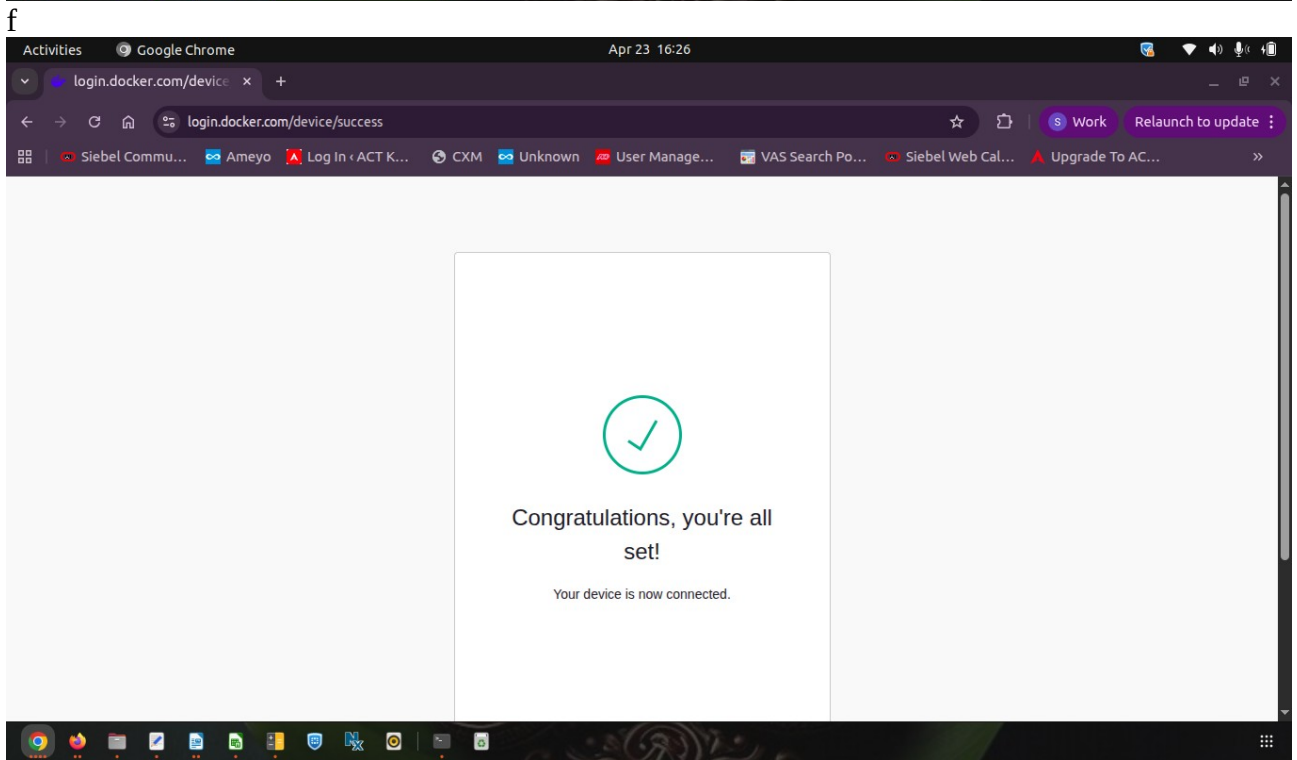
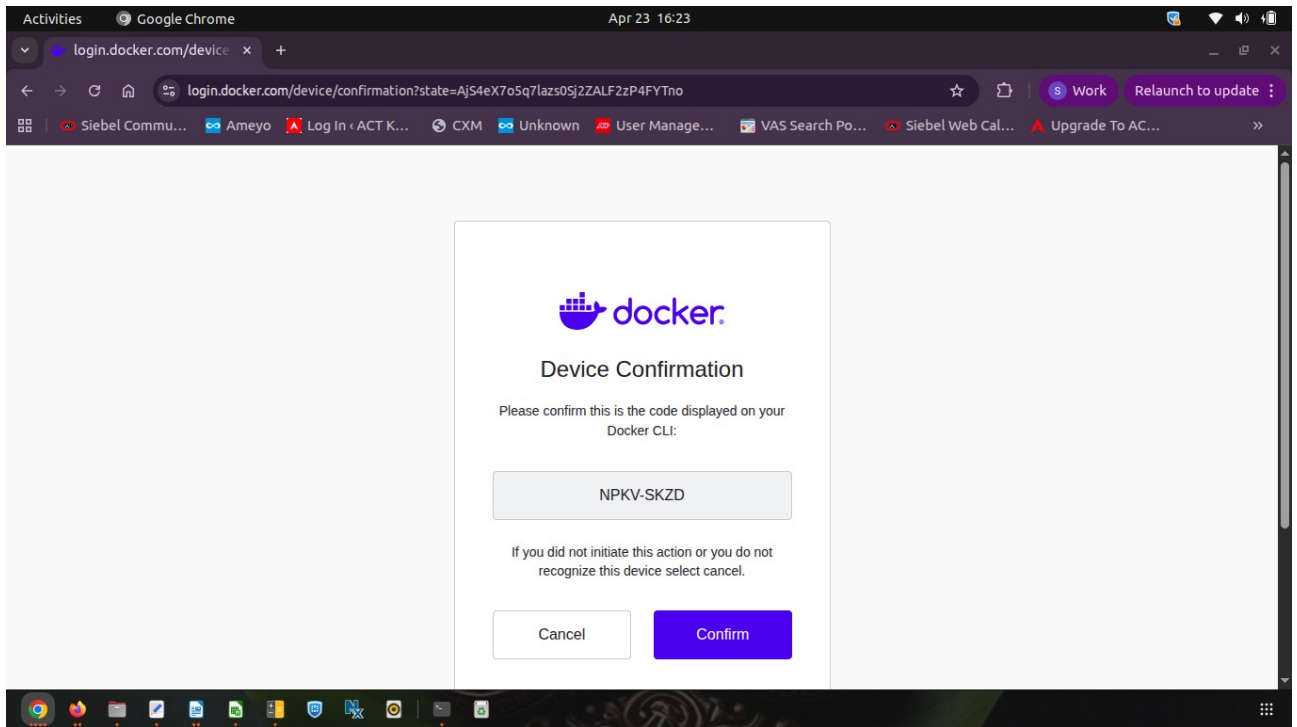
PFA Token



d

```
docker login -u 30727
```

```
dckr_pat_JDIyCcayU3xnNPK2n5szvVnbs4g
```

S

docker logged via sudo

```
ubuntu@ip-172-31-46-126: ~/devops-build
Waiting for authentication in the browser...

https://login.docker.com/activate
failed waiting for authentication: timed out waiting for device token
ubuntu@ip-172-31-46-126:~/devops-build$ sudo docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: GSPC-FJFZ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

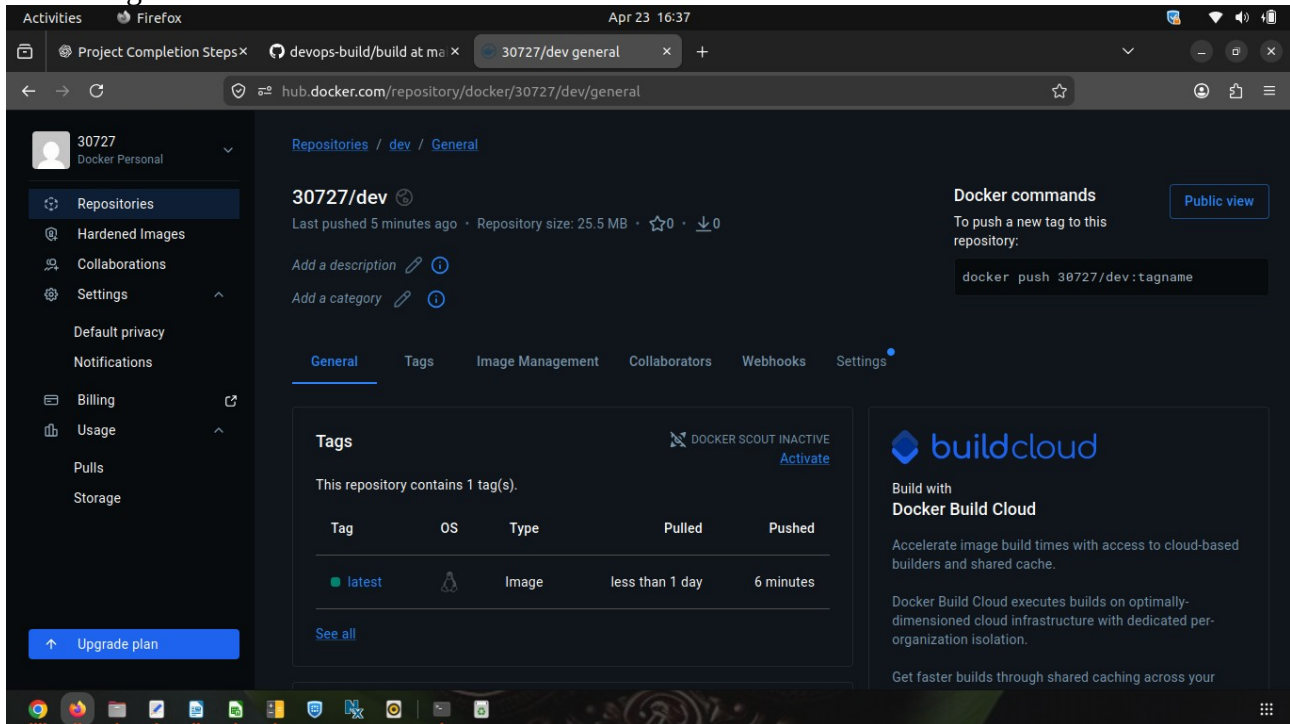
Login Succeeded
ubuntu@ip-172-31-46-126:~/devops-build$
```

d

```
ubuntu@ip-172-31-46-126: ~/devops-build
47c0a68c4b82: Waiting
aee4e54b3865: Waiting
6a0ac1617861: Waiting
583599bb7d38: Waiting
781ff50d2644: Waiting
push access denied, repository does not exist or may require authorization: server message: insufficient_scope: authorization failed
ubuntu@ip-172-31-46-126:~/devops-build$ sudo docker push 30727/dev:latest
The push refers to repository [docker.io/30727/dev]
612c0c1df4c5: Mounted from library/nginx
47c0a68c4b82: Pushed
82736a35d0e7: Mounted from library/nginx
583599bb7d38: Mounted from library/nginx
edce8baeff07: Pushed
aee4e54b3865: Mounted from library/nginx
6a0ac1617861: Mounted from library/nginx
781ff50d2644: Mounted from library/nginx
453da7dbc73e: Mounted from library/nginx
4a8b0b2a5b19: Mounted from library/nginx
latest: digest: sha256:a6742b90955f671b6d91fb737c10e01ece21a159c360f0eee8899217c1fb4bde size: 2478
ubuntu@ip-172-31-46-126:~/devops-build$ docker logout
docker login -u 30727
Removing login credentials for https://index.docker.io/v1/
```

d

Push Images to docker



Activities Firefox Apr 23 16:37

Project Completion Steps x devops-build/build at ma x 30727/dev general x +

hub.docker.com/repository/docker/30727/dev/general

30727 Docker Personal

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Upgrade plan

Repositories / dev / General

30727/dev

Last pushed 5 minutes ago · Repository size: 25.5 MB · ☆0 · ↓0

Add a description

Add a category

Docker commands

To push a new tag to this repository:

```
docker push 30727/dev:tagname

Public view



General Tags Image Management Collaborators Webhooks Settings



Tags



This repository contains 1 tag(s).



| Tag    | OS    | Type  | Pulled          | Pushed    |
|--------|-------|-------|-----------------|-----------|
| latest | linux | Image | less than 1 day | 6 minutes |



See all



DOCKER SCOUT INACTIVE



Activate



Build with Docker Build Cloud



Accelerate image build times with access to cloud-based builders and shared cache.



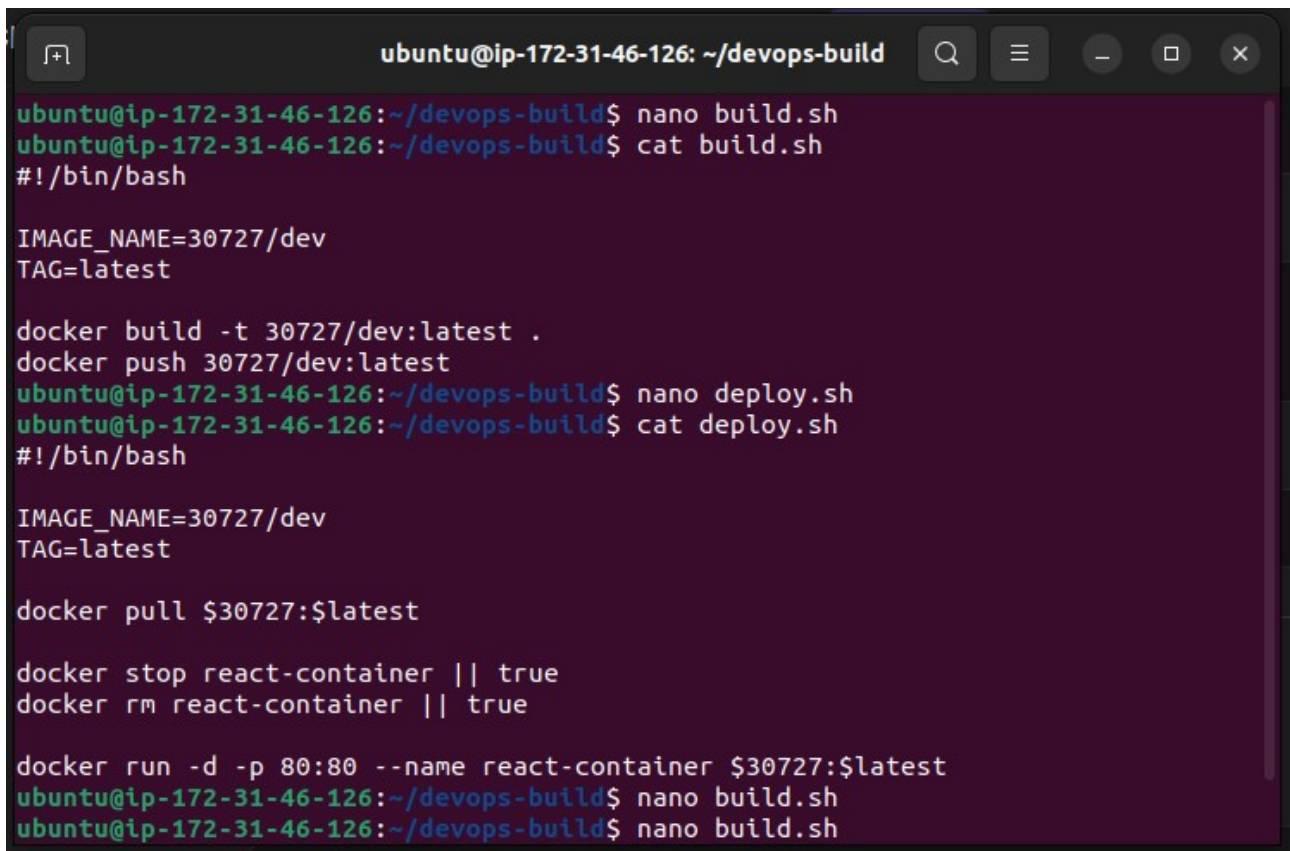
Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.



Get faster builds through shared caching across your


```

d
Bash Scripts



```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ nano build.sh
ubuntu@ip-172-31-46-126:~/devops-build$ cat build.sh
#!/bin/bash

IMAGE_NAME=30727/dev
TAG=latest

docker build -t 30727/dev:latest .
docker push 30727/dev:latest
ubuntu@ip-172-31-46-126:~/devops-build$ nano deploy.sh
ubuntu@ip-172-31-46-126:~/devops-build$ cat deploy.sh
#!/bin/bash

IMAGE_NAME=30727/dev
TAG=latest

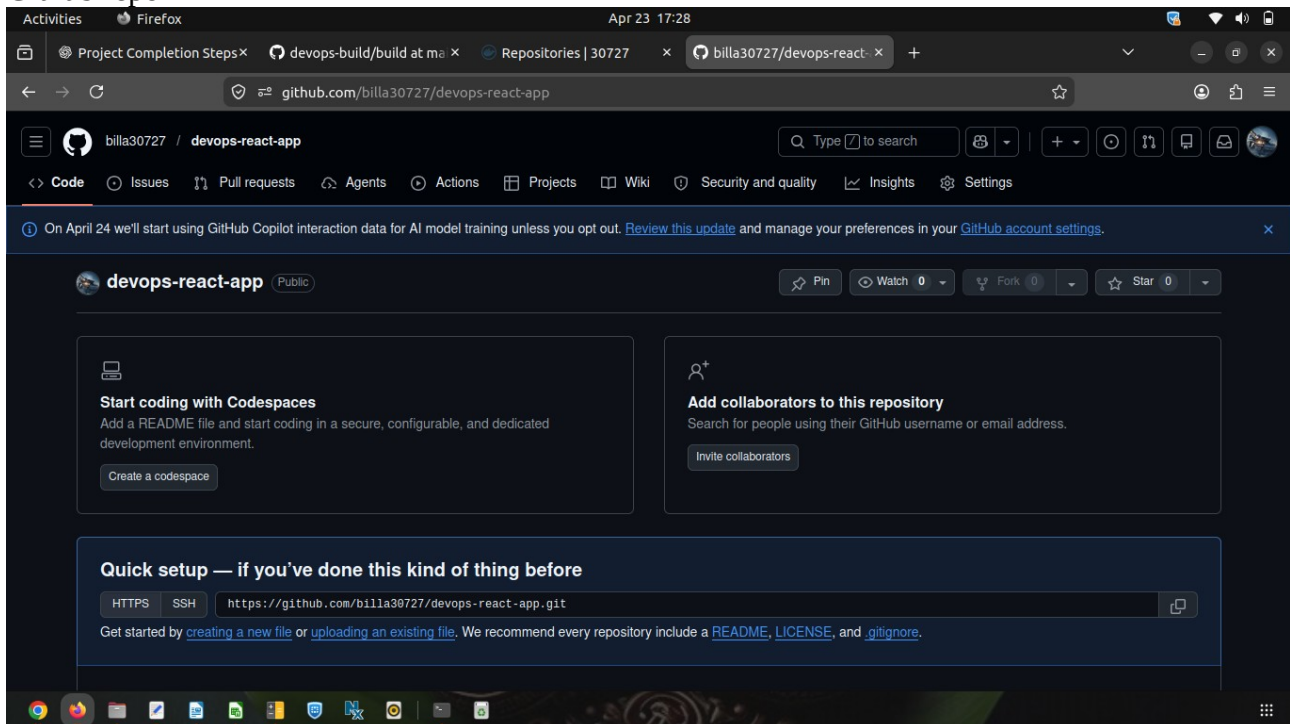
docker pull $30727:$latest

docker stop react-container || true
docker rm react-container || true

docker run -d -p 80:80 --name react-container $30727:$latest
ubuntu@ip-172-31-46-126:~/devops-build$ nano build.sh
ubuntu@ip-172-31-46-126:~/devops-build$ nano build.sh
```

f

Github repo



Install Java

```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ sudo apt install openjdk-17-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common
  at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service
  fontconfig fonts-dejavu-extra gsettings-desktop-schemas
  gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java
  libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-client3
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2
  libcups2t64 libdatrie1 libdconf1 libgail-common libgail18t64 libgif7
  libgraphite2-3 libgtk2.0-0t64 libgtk2.0-bin libgtk2.0-common libharfbuzz0b
  libice-dev liblcms2-2 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0
  libpcsclite1 libpixmap-1-0 libpthread-stubs0-dev librsvg2-2 librsvg2-common
  libsm-dev libthai-data libthai0 libx11-dev libxau-dev libxcb-render0
  libxcb1-dev libxdamage1 libxdmcp-dev libxt-dev openjdk-17-jdk-headless
  openjdk-17-jre openjdk-17-jre-headless session-migration ubuntu-mono
  x11proto-dev xorg-sgml-doctools xtrans-dev
Suggested packages:
  default-jre alsa-utils libasound2-plugins cups-common gvfs libice-doc
  liblcms2-utils pscd librsvg2-bin libsm-doc libx11-doc libxcb-doc libxt-doc
  openjdk-17-demo openjdk-17-source visualvm libnss-mdns fonts-ipafont-gothic
```

f


```
ubuntu@ip-172-31-46-126: ~/devops-build
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up openjdk-17-jdk:amd64 (17.0.18+8-1~24.04.1) ...
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart systemd-logind.service

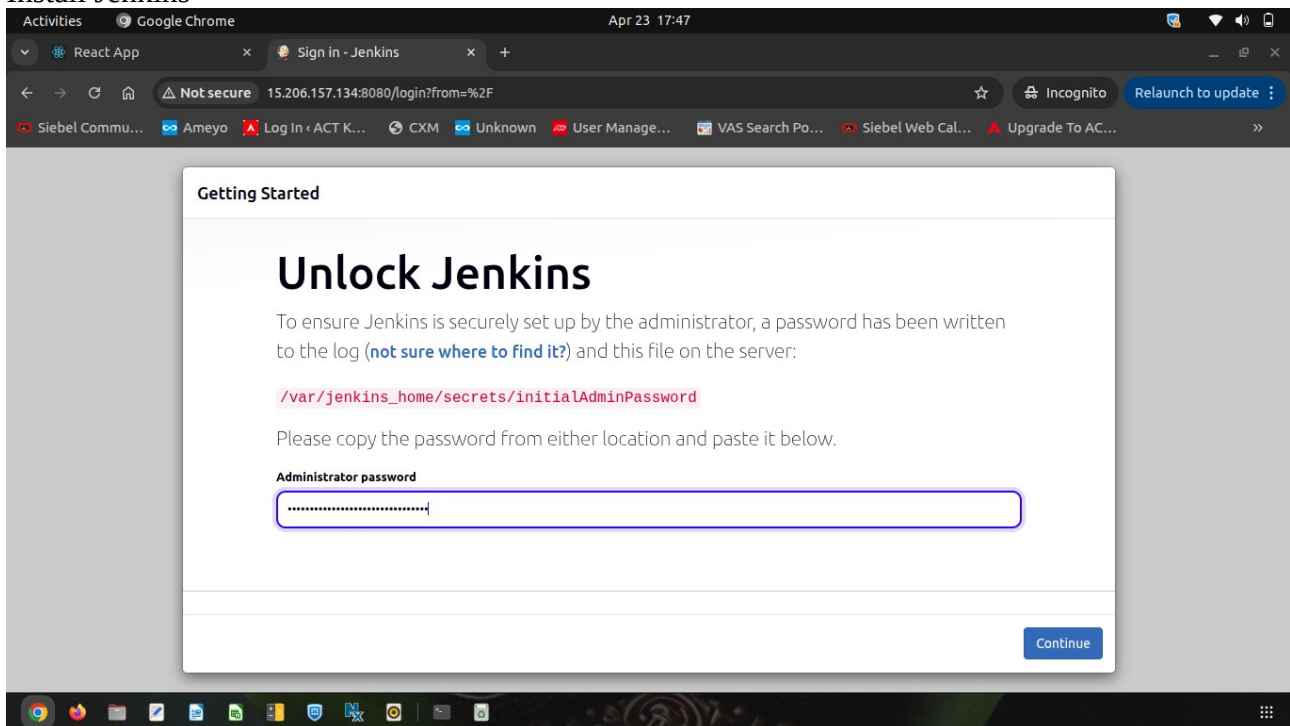
No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #3: sshd[1081]
ubuntu @ user manager service: systemd[1086]

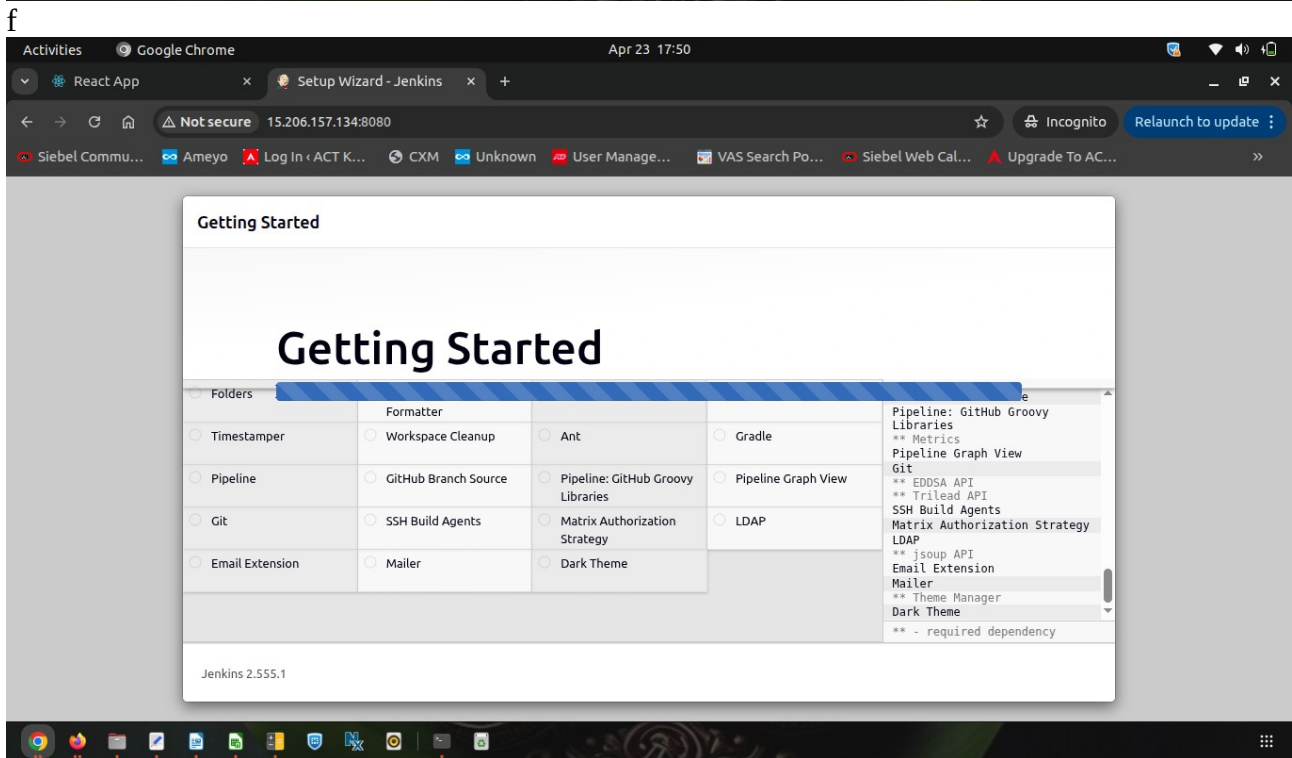
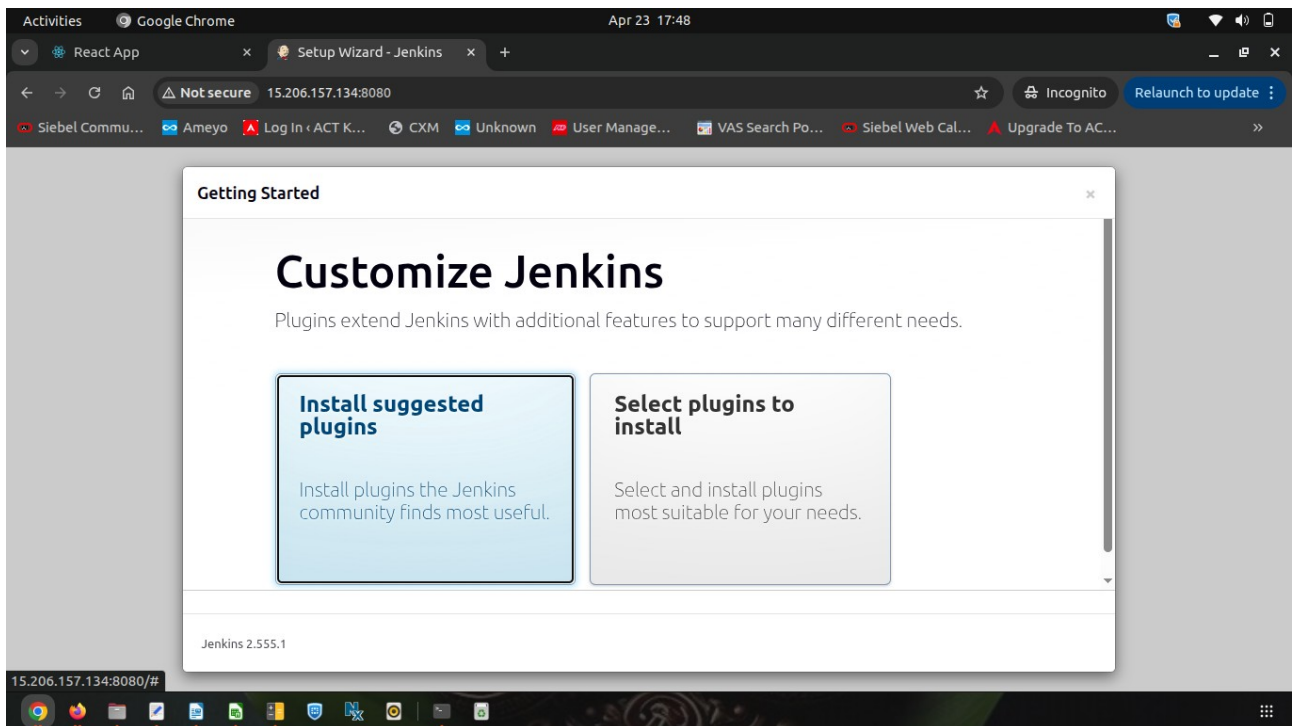
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-46-126:~/devops-build$
```

i

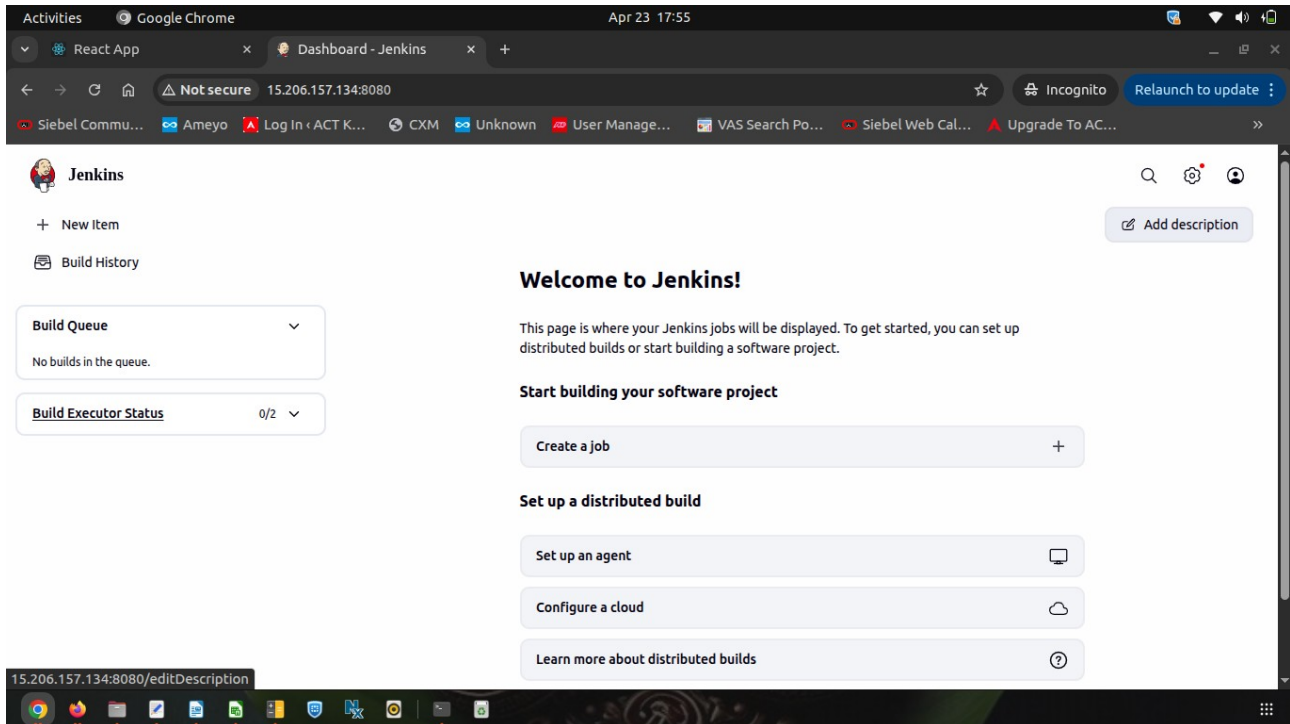
Install Jenkins



f



Create Jenkins Job



The screenshot shows the Jenkins Dashboard in a Google Chrome browser window. The address bar indicates the URL is 15.206.157.134:8080. The dashboard features a sidebar with 'New Item' and 'Build History' links. The main content area has a 'Welcome to Jenkins!' message and a 'Start building your software project' section with buttons for 'Create a job', 'Set up a distributed build', 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. A 'Build Queue' widget shows 'No builds in the queue' and a 'Build Executor Status' widget shows '0/2'.

Activities Google Chrome Apr 23 17:55

React App Dashboard - Jenkins

Not secure 15.206.157.134:8080

Siebel Commu... Ameyo Log In < ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status 0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job +

Set up a distributed build

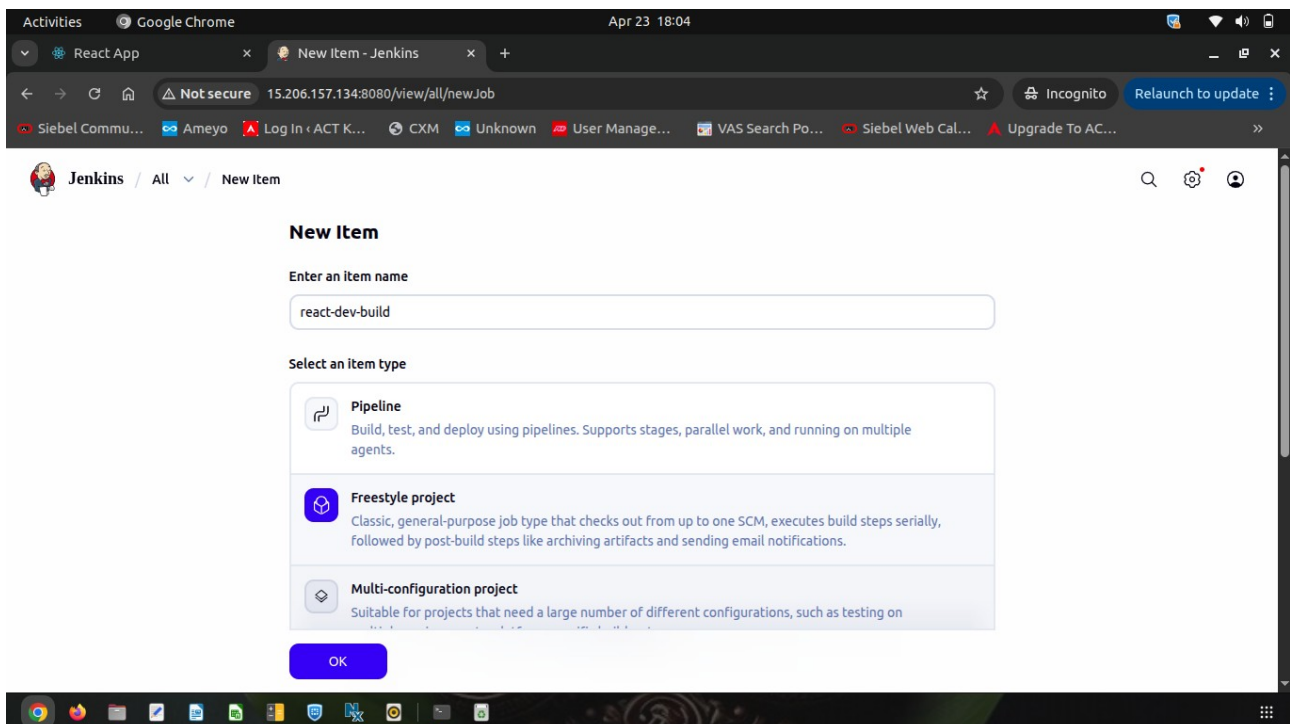
Set up an agent

Configure a cloud

Learn more about distributed builds

15.206.157.134:8080/editDescription

f



The screenshot shows the 'New Item' form in the Jenkins browser interface. The 'Enter an item name' field contains 'react-dev-build'. Under 'Select an item type', three options are listed: 'Pipeline' (Build, test, and deploy using pipelines), 'Freestyle project' (Classic, general-purpose job type), and 'Multi-configuration project' (Suitable for projects that need a large number of different configurations). An 'OK' button is at the bottom.

Activities Google Chrome Apr 23 18:04

React App New Item - Jenkins

Not secure 15.206.157.134:8080/view/all/newJob

Siebel Commu... Ameyo Log In < ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / All / New Item

New Item

Enter an item name

react-dev-build

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on

OK

f

Activities Google Chrome Apr 23 18:14

React App react-dev-build Config - x +

Not secure 15.206.157.134:8080/job/react-dev-build/configure

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled

Description

Plain text [Preview](#)

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterised ?

☐ Throttle builds ?

[Save](#) [Apply](#)

Activities Google Chrome Apr 23 18:15

React App react-dev-build Config - x +

Not secure 15.206.157.134:8080/job/react-dev-build/configure

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

<https://github.com/billa30727/devops-react-app.git>

Credentials ?

- none - [+ Add](#)

Advanced [v](#)

[Save](#) [Apply](#)

Activities Google Chrome Apr 23 18:15

React App react-dev-build Config - x +

Not secure 15.206.157.134:8080/job/react-dev-build/configure

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Branches to build ?

Branch Specifier (blank for 'any') ?

*/dev

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Save Apply

Activities Google Chrome Apr 23 18:15

React App react-dev-build Config - x +

Not secure 15.206.157.134:8080/job/react-dev-build/configure

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☐ GitHub Branches
- ☐ GitHub Pull Requests ?
- ☒ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?

Environment

Configure settings and variables that define the context in which your build runs. Use predefined settings and global parameters.

Save Apply

Activities Google Chrome Apr 23 18:15

React App react-dev-build Config - x +

Not secure 15.206.157.134:8080/job/react-dev-build/configure

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

```
chmod +x build.sh
./build.sh
```

Advanced ^

Exit code to set build unstable ?

Save Apply

f

Activities Google Chrome Apr 23 18:15

React App react-dev-build Config - x +

Not secure 15.206.157.134:8080/job/react-dev-build/configure

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions**

+ Add environment filter

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

+ Add post-build action

Save Apply

REST API Jenkins 2.555.1

f

Install the docker inside the jenkins container

```
ubuntu@ip-172-31-46-126: ~/devops-build
exit
ubuntu@ip-172-31-46-126:~/devops-build$ docker exec -u root -it jenkins bash
root@b6bf9396277c:/# apt update
apt install docker.io -y
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages
[127 kB]
Fetched 10.0 MB in 1s (8830 kB/s)
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  docker.io

Installing dependencies:
  apparmor          libintl-xs-perl      libsort-naturally-perl
  containerd        libip4tc2            libsystemd-shared
  criu              libip6tc2            libterm-readkey-perl
  dbus              libjansson4          libtirpc-common
  dbus-bin          libjson-c5           libtirpc3t64
  dbus-daemon       libkmod2             libxtables12
  dbus-session-bus-common libmnl0              linux-sysctl-defaults
```

```
f
Installing dependencies:
  apparmor          libintl-xs-perl      libsort-naturally-perl
  containerd        libip4tc2            libsystemd-shared
  criu              libip6tc2            libterm-readkey-perl
  dbus              libjansson4          libtirpc-common
  dbus-bin          libjson-c5           libtirpc3t64
  dbus-daemon       libkmod2             libxtables12
  dbus-session-bus-common libmnl0              linux-sysctl-defaults
  dbus-system-bus-common libmodule-find-perl  media-types
  dbus-user-session libnet1              needrestart
  dmsetup           libnetfilter-conntrack3 netbase
  docker-buildx     libnfnetlink0        nftables
  docker-cli        libnftables1         python3
  gettext-base     libnftnl11           python3-minimal
  iproute2          libnl-3-200          python3-protobuf
  iptables          libnss-systemd       python3-pycriu
  libapparmor1      libpam-cap           python3.13
  libbbpf1          libpam-systemd       python3.13-minimal
  libcap2-bin       libproc-processtable-perl readline-common
  libcompel1        libprotobuf-c1       runc
  libcryptsetup12   libprotobuf32t64     systemd
  libdbus-1-3       libpython3-stdlib    systemd-cryptsetup
  libdevmapper1.02.1 libpython3.13-minimal systemd-sysv
```

f

```
ubuntu@ip-172-31-46-126: ~/devops-build
Suggested packages:
  apparmor-profiles-extra      firewallld      python3.13-venv
  apparmor-utils              kmod            python3.13-doc
  containernetworking-plugins libtss2-rc0t64  binutils
  docker-doc                  libarchive13t64 bintfmt-support
  aufs-tools                  libdw1t64       readline-doc
  btrfs-progs                 libgcrypt20     systemd-container
  cgroupfs-mount              libpwquality1   systemd-homed
  debootstrap                 libgrencode4    systemd-userdbd
  e2fsprogs                   needrestart-session systemd-boot
  rinse                       | libnotify-bin systemd-resolved
  rootlesskit                 iucode-tool     systemd-repart
  xfsprogs                    python3-doc     libtss2-tcti-device0
  zfs-fuse                    python3-tk       polkitd
  | zfsutils-linux            python3-venv

Summary:
  Upgrading: 0, Installing: 73, Removing: 0, Not Upgrading: 2
  Download size: 100 MB
  Space needed: 401 MB / 44.3 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 libsystemd-shared amd64 257
.9-1~deb13u1 [2151 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libapparmor1 amd64 4.1.0-1
```

f

```
ubuntu@ip-172-31-46-126: ~/devops-build
sysctl: permission denied on key "kernel.core_pattern"
sysctl: permission denied on key "kernel.sysrq"
sysctl: permission denied on key "kernel.core_uses_pid"
sysctl: permission denied on key "net.ipv4.conf.default.rp_filter"
sysctl: permission denied on key "net.ipv4.conf.eth0.rp_filter"
sysctl: permission denied on key "net.ipv4.conf.lo.rp_filter"
sysctl: permission denied on key "net.ipv4.conf.default.accept_source_route"
sysctl: permission denied on key "net.ipv4.conf.eth0.accept_source_route"
sysctl: permission denied on key "net.ipv4.conf.lo.accept_source_route"
sysctl: permission denied on key "net.ipv4.conf.default.promote_secondaries"
sysctl: permission denied on key "net.ipv4.conf.eth0.promote_secondaries"
sysctl: permission denied on key "net.ipv4.conf.lo.promote_secondaries"
sysctl: permission denied on key "net.ipv4.ping_group_range", ignoring
sysctl: permission denied on key "fs.protected_hardlinks"
sysctl: permission denied on key "fs.protected_symlinks"
sysctl: permission denied on key "fs.protected_regular"
sysctl: permission denied on key "fs.protected_fifos"
sysctl: permission denied on key "vm.max_map_count"
sysctl: permission denied on key "kernel.pid_max"
Processing triggers for libc-bin (2.41-12+deb13u2) ...
Processing triggers for systemd (257.9-1~deb13u1) ...
root@b6bf9396277c:/#
```

f

deploy.sh

```
ubuntu@ip-172-31-46-126: ~/devops-build
^Z
[1]+  Stopped                  ping github.com
ubuntu@ip-172-31-46-126:~/devops-build$ git push origin dev
Username for 'https://github.com': billa30727
Password for 'https://billa30727@github.com':
Everything up-to-date
ubuntu@ip-172-31-46-126:~/devops-build$ nano deploy.sh
ubuntu@ip-172-31-46-126:~/devops-build$ cat deploy.sh
#!/bin/bash

IMAGE_NAME="30727/dev:latest"

echo "Stopping old container..."
docker stop react-app || true
docker rm react-app || true

echo "Pulling latest image..."
docker pull $IMAGE_NAME

echo "Running container on port 80..."
docker run -d -p 80:80 --name react-app $IMAGE_NAME

echo "Deployment successful 🚀"
ubuntu@ip-172-31-46-126:~/devops-build$
```

f

```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED
STATUS        PORTS                               NAMES
a88abc66a03a   louislam/uptime-kuma               "/usr/bin/dumb-init ..." About an hour ago
Up About an hour (healthy)   0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp
                                uptime-kuma
b6bf9396277c   jenkins/jenkins:lts               "/usr/bin/tini -- /u..." 4 hours ago
Up 4 hours                  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0
.0.0:50000->50000/tcp, [::]:50000->50000/tcp   jenkins
ubuntu@ip-172-31-46-126:~/devops-build$ ./deploy.sh
Stopping old container...
react-app
react-app
Pulling latest image...
latest: Pulling from 30727/dev
Digest: sha256:cdebbd1ebe0e9c9d21025b4e89227a53df25cbc7aba840c3125cc3c6e1c2033e
Status: Image is up to date for 30727/dev:latest
docker.io/30727/dev:latest
Running container...
d7881c234ee9df0f24b1324fc087d576eb5096d2d2fe55c600a9e5ab99b75efb
Deployment SUCCESS 🚀
ubuntu@ip-172-31-46-126:~/devops-build$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED
```

f

```
ubuntu@ip-172-31-46-126: ~/devops-build
Up 4 hours                                0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp jenkins
ubuntu@ip-172-31-46-126:~/devops-build$ curl ifconfig.me
15.206.157.134ubuntu@ip-172-31-46-126:~/devops-build$ cat build.sh
#!/bin/bash

set -e

# Correct repo
REPO="30727/dev"

# Tag
TAG=$(date +%s)

echo "Building image..."
docker build -t $REPO:$TAG .

echo "Tagging latest..."
docker tag $REPO:$TAG $REPO:latest

echo "Login to Docker Hub..."
echo "Admin@12345" | docker login -u 30727 --password-stdin
```

f

```
ubuntu@ip-172-31-46-126: ~/devops-build
docker push $REPO:latest
echo "SUCCESS 🚀"
ubuntu@ip-172-31-46-126:~/devops-build$ cat deploy.sh
#!/bin/bash

set -e

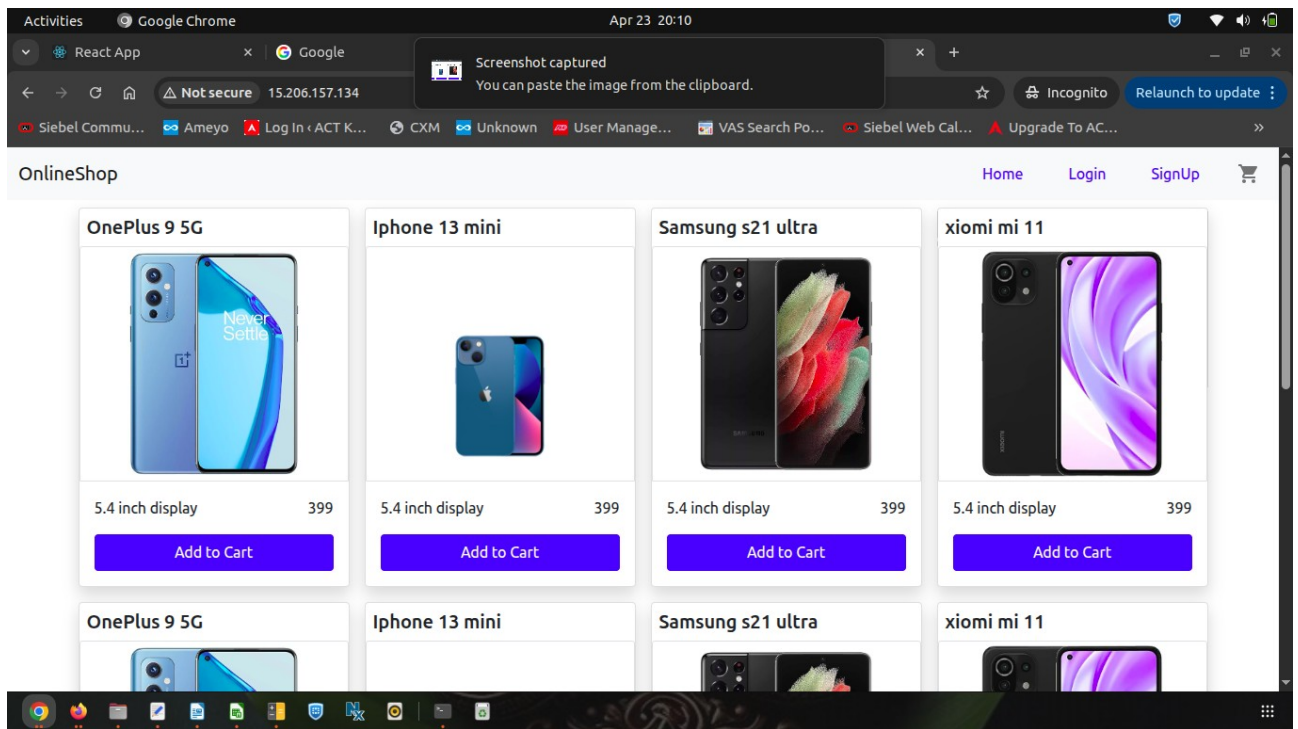
IMAGE="30727/dev:latest"

echo "Stopping old container..."
docker stop react-app || true
docker rm react-app || true

echo "Pulling latest image..."
docker pull $IMAGE

echo "Running container..."
docker run -d -p 80:80 --name react-app $IMAGE

echo "Deployment SUCCESS 🚀"
ubuntu@ip-172-31-46-126:~/devops-build$
```



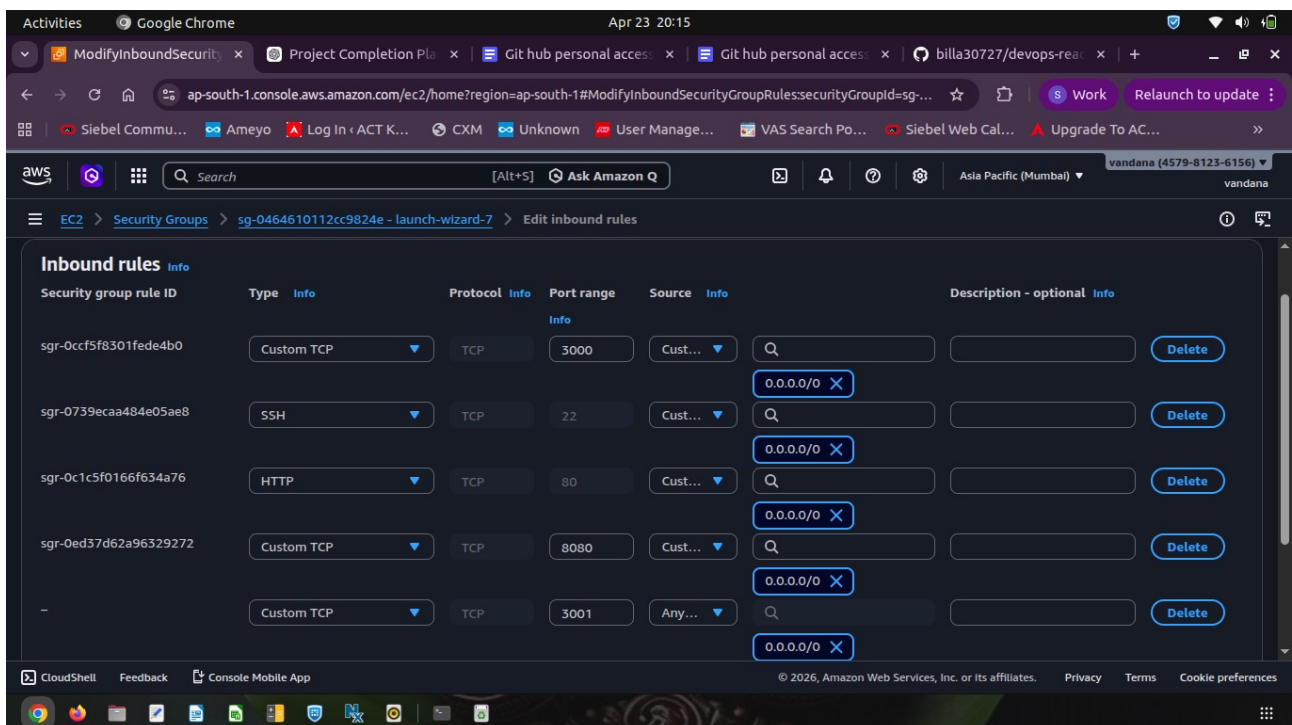
f

Monitoring Setup

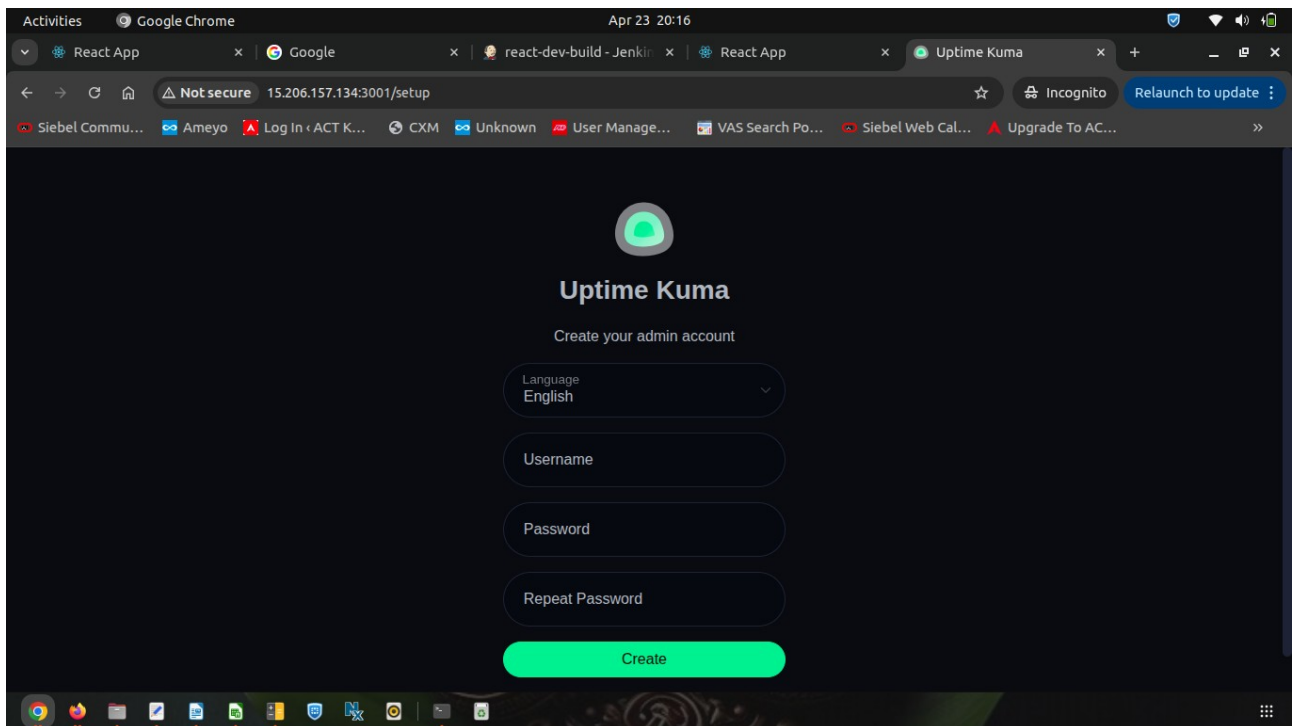
```
ubuntu@ip-172-31-46-126: ~/devops-build
ubuntu@ip-172-31-46-126:~/devops-build$ docker run -d \
  --name uptime-kuma \
  -p 3001:3001 \
  louislam/uptime-kuma
Unable to find image 'louislam/uptime-kuma:latest' locally
latest: Pulling from louislam/uptime-kuma
7d955db85b85: Pull complete
ab019b589bd1: Pull complete
05489cad7e07: Pull complete
874bf4d93720: Pull complete
2da5495de763: Pull complete
0009b3591d8d: Pull complete
b16337721583: Pull complete
b5cfa7561d78: Pull complete
ad1253342b11: Pull complete
7607ceb74d22: Pull complete
4f4fb700ef54: Pull complete
b338562f40a7: Pull complete
2c706596bd17: Pull complete
0dc64e6ca98d: Download complete
Digest: sha256:3d632903e6af34139a37f18055c4f1bfd9b7205ae1138f1e5e8940ddc1d176f9
Status: Downloaded newer image for louislam/uptime-kuma:latest
a88abc66a03a94ad9ceab223e881ba018e25fae6703c6ad6b33fe02061486249
ubuntu@ip-172-31-46-126:~/devops-build$
```

f

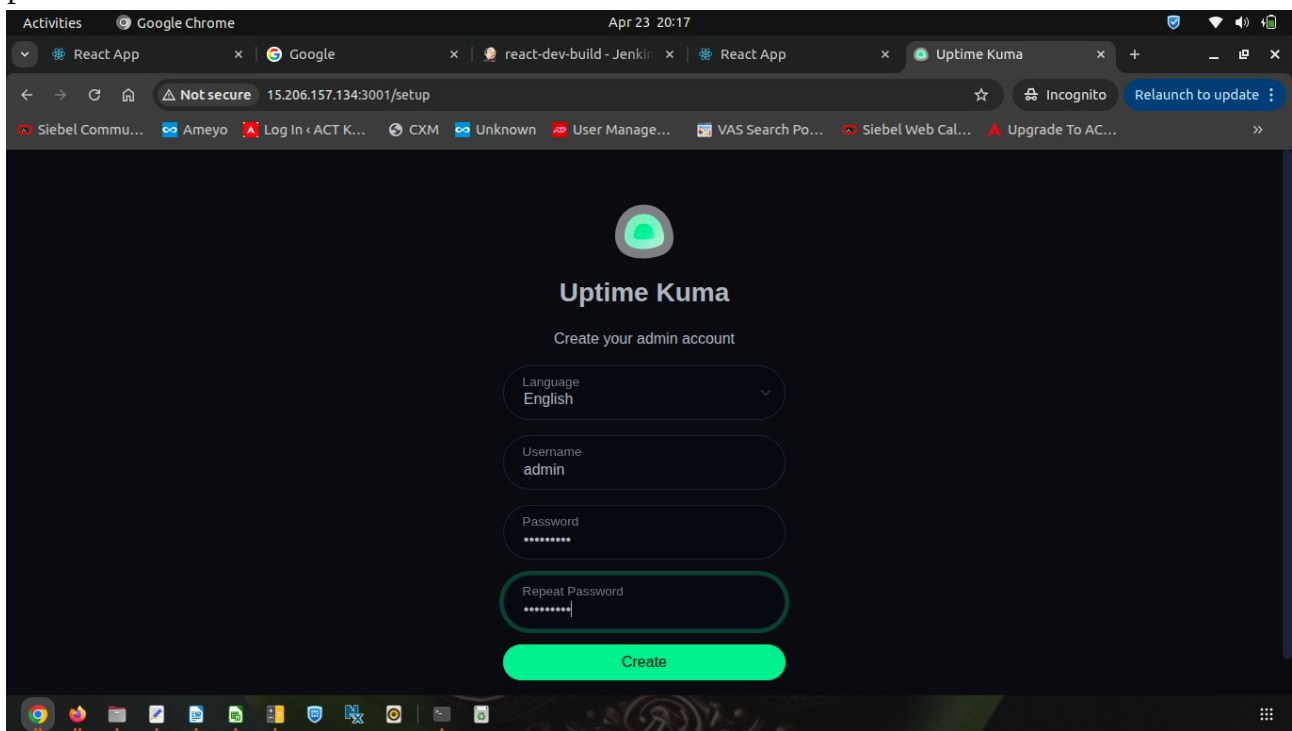
Add the inbound rules monitoring tool run



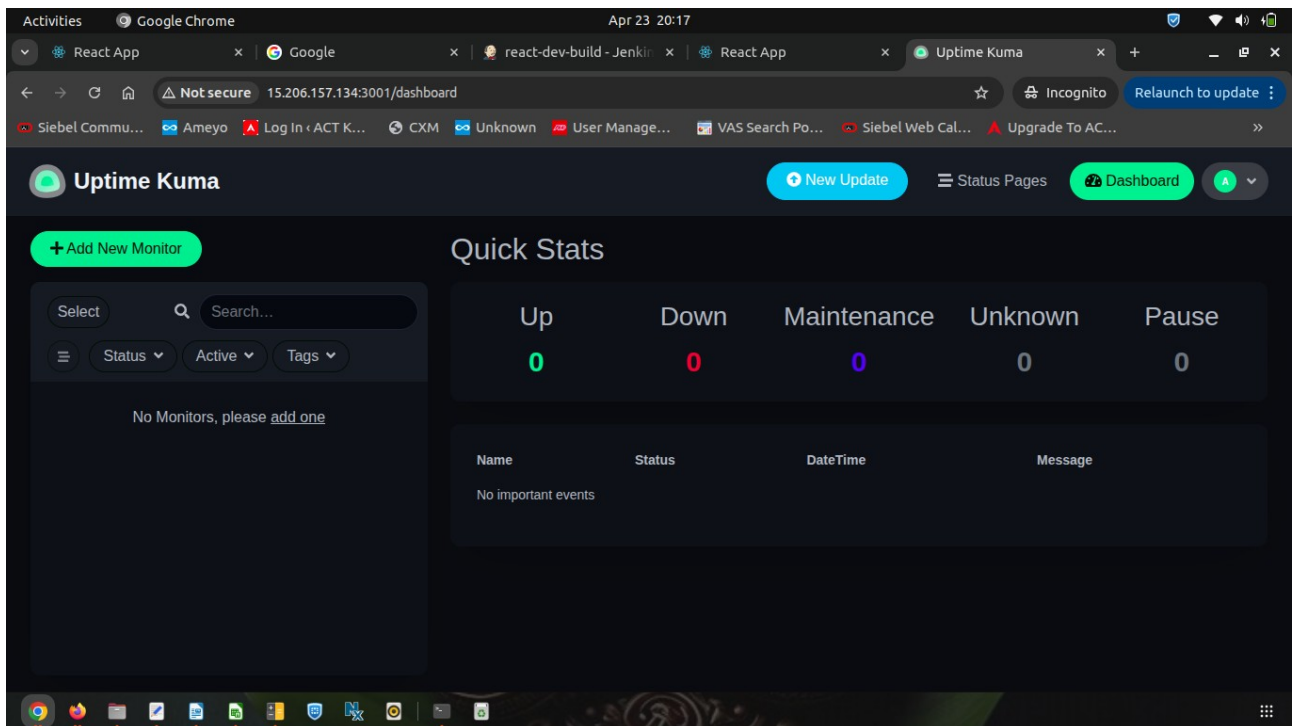
f



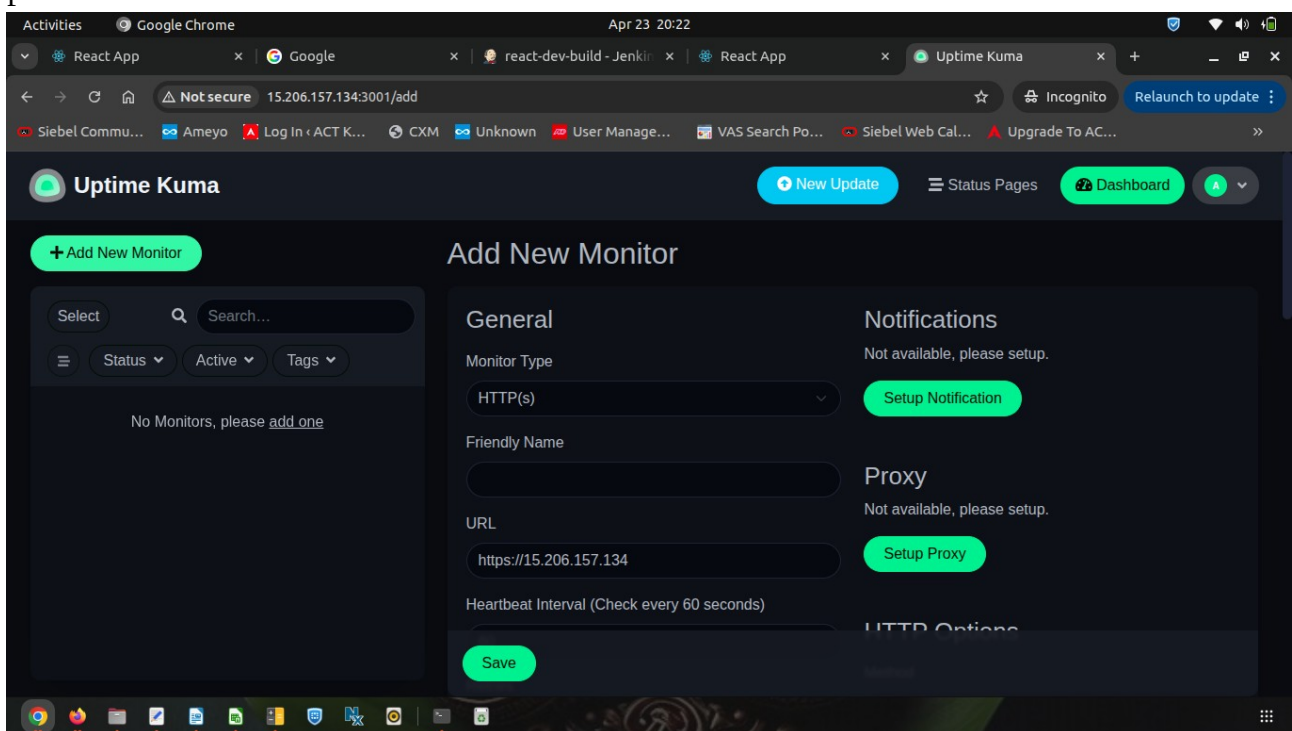
f



f



f



f

Activities Google Chrome Apr 23 20:27

React App x Google x react-dev-build - Jenkins x React App x Uptime Kuma x + -

Not secure 15.206.157.134:3001/dashboard/1 ☆ Incognito Relaunch to update

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Uptime Kuma New Update Status Pages Dashboard A

+ Add New Monitor

Select Search... Status Active Tags

N/A devops-build

devops-build <https://15.206.157.134/> Pause Edit Clone Delete

Unknown

Check every 60 seconds

Response	Avg. Response	Uptime	Uptime
(Current)	(24-hour)	(24-hour)	(30-day)
N/A	N/A	N/A	N/A

f

Activities Google Chrome Apr 23 20:40

React App x Google x Console - #9 - Jenkins x Uptime Kuma x React App x + -

Not secure 15.206.157.134:3001/dashboard/1 ☆ Incognito Relaunch to update

Siebel Commu... Ameyo Log In ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Uptime Kuma New Update Status Pages Dashboard A

+ Add New Monitor

Select Search... Status Active Tags

0% devops-build

devops-build <https://15.206.157.134/> Pause Edit Clone Delete

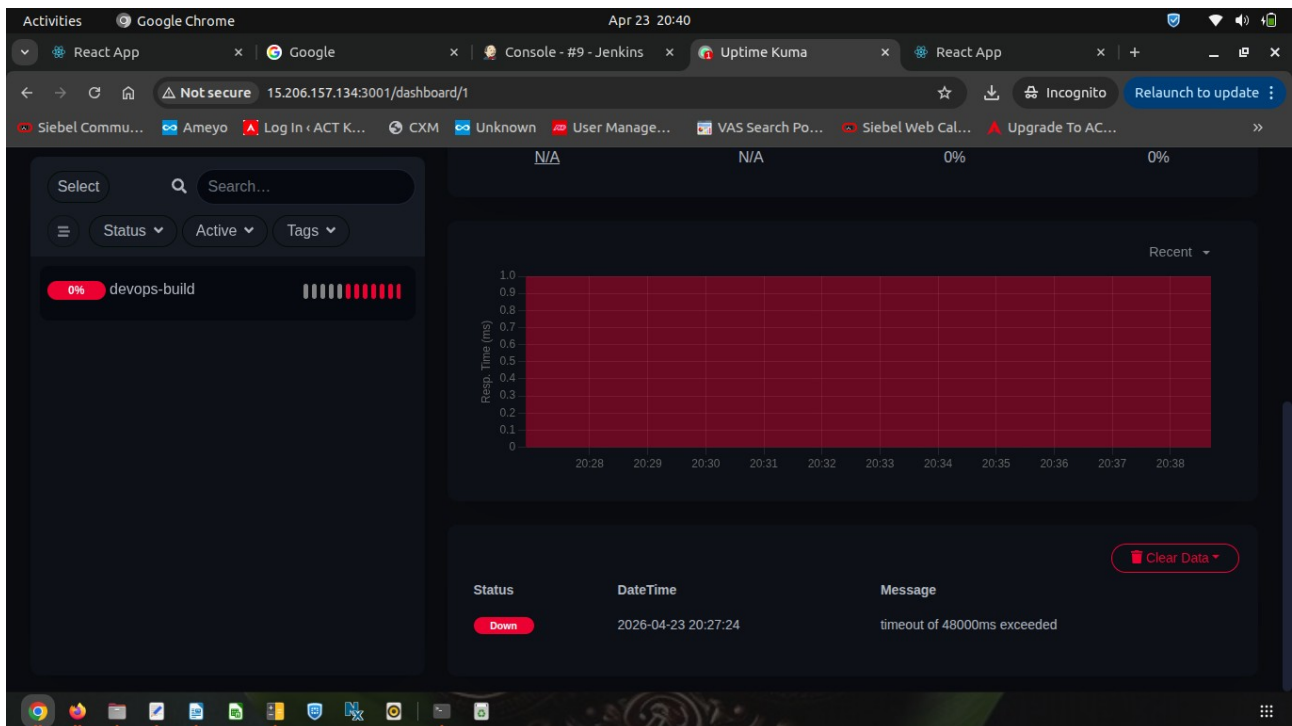
Down

11m ago now

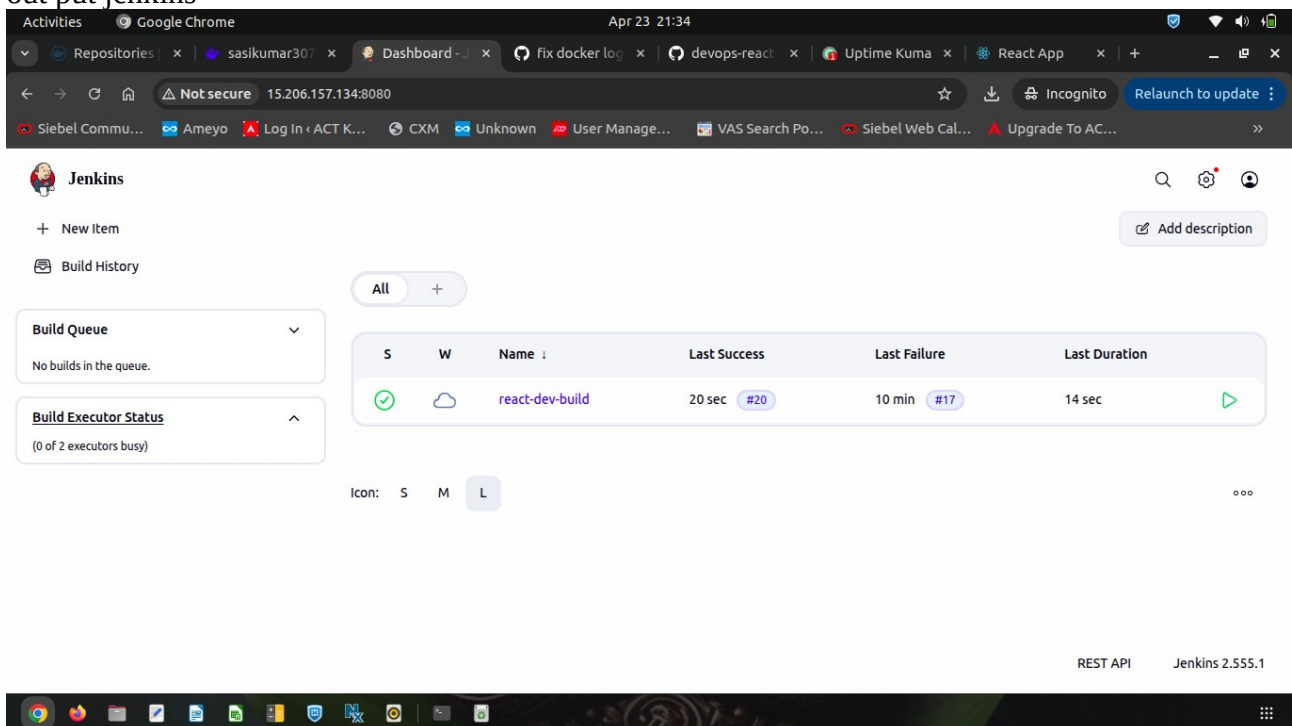
Check every 60 seconds

Response	Avg. Response	Uptime	Uptime
(Current)	(24-hour)	(24-hour)	(30-day)
N/A	N/A	0%	0%

f



C out put jenkins



f

Activities Google Chrome Apr 23 21:32

Repositories x sasikumar307 x Console - #18 x fix docker log x devops-react x Uptime Kuma x React App x +

Not secure 15.206.157.134:8080/job/react-dev-build/18/console Relaunch to update

Siebel Commu... Ameyo Log In < ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / #18

Status

</> Changes

Console Output

Edit Build Information

Delete build '#18'

Timings

Git Build Data

Previous Build

Console

Download Copy View as plain text

```
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/react-dev-build
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/react-dev-build/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/billa30727/devops-react-app.git # timeout=10
Fetching upstream changes from https://github.com/billa30727/devops-react-app.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/billa30727/devops-react-app.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/dev^{commit} # timeout=10
Checking out Revision 4bcda7589cf4e6ed35ee713388cedfc9cac0bbc0 (refs/remotes/origin/dev)
> git config core.sparsecheckout # timeout=10
> git checkout -f 4bcda7589cf4e6ed35ee713388cedfc9cac0bbc0 # timeout=10
Commit message: "fix deploy script"
> git rev-list --no-walk 4bcda7589cf4e6ed35ee713388cedfc9cac0bbc0 # timeout=10
```

f

Activities Google Chrome Apr 23 21:33

Repositories x sasikumar307 x Console - #18 x fix docker log x devops-react x Uptime Kuma x React App x +

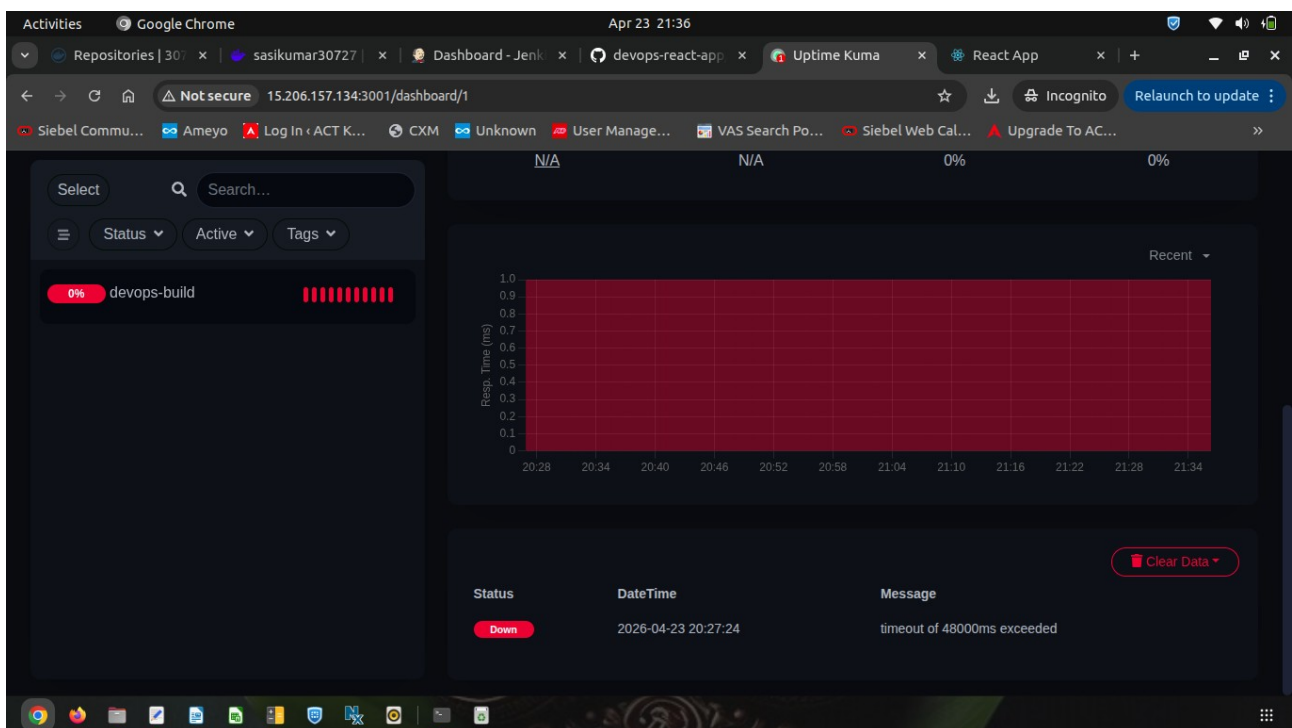
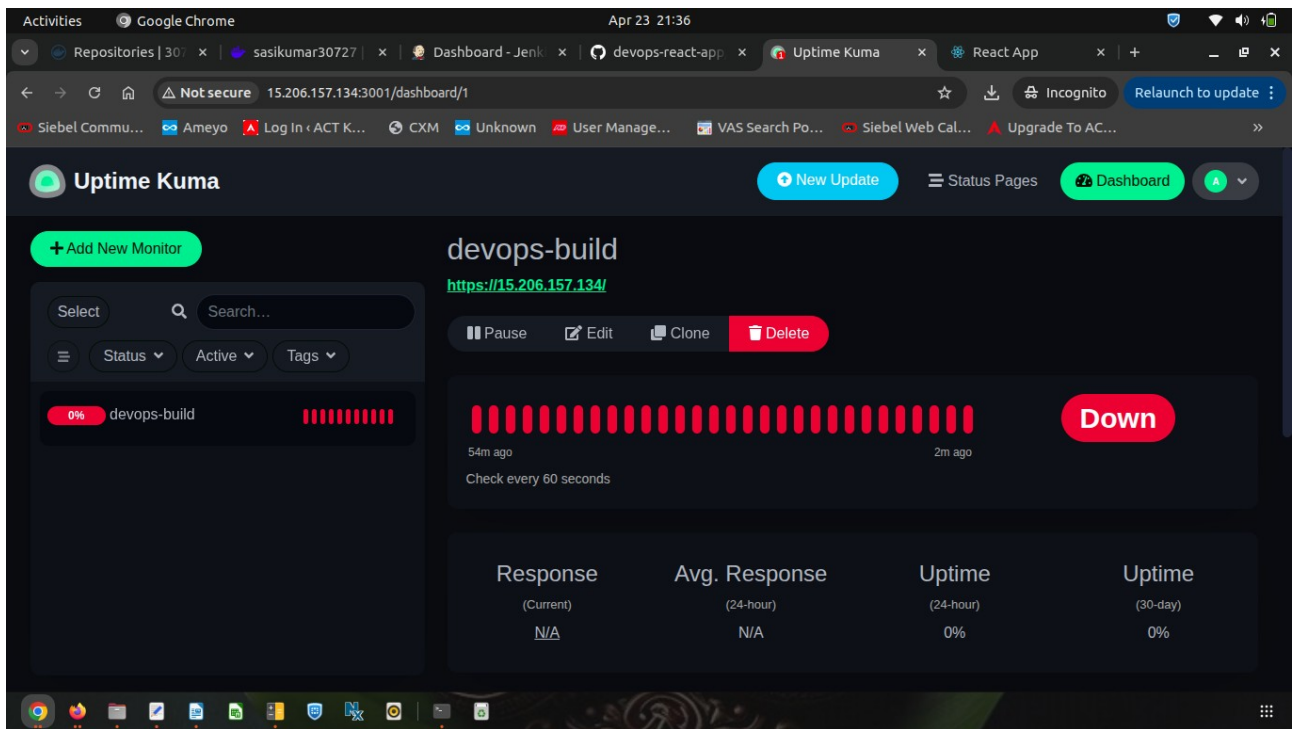
Not secure 15.206.157.134:8080/job/react-dev-build/18/console Relaunch to update

Siebel Commu... Ameyo Log In < ACT K... CXM Unknown User Manage... VAS Search Po... Siebel Web Cal... Upgrade To AC...

Jenkins / react-dev-build / #18

```
82736a35d0e7: waiting
82736a35d0e7: Waiting
82736a35d0e7: Waiting
82736a35d0e7: Waiting
82736a35d0e7: Layer already exists
latest: digest: sha256:000331065f43bb8f8e0a1c51719349814a0a11d8b1f60a32d4bc493fcac0dc15 size: 856
SUCCESS
[react-dev-build] $ /bin/sh -xe /tmp/jenkins2558364451940689159.sh
+ chmod +x deploy.sh
+ ./deploy.sh
Stopping old container...
react-app
react-app
Pulling latest image...
latest: Pulling from 30727/dev
Digest: sha256:000331065f43bb8f8e0a1c51719349814a0a11d8b1f60a32d4bc493fcac0dc15
Status: Image is up to date for 30727/dev:latest
docker.io/30727/dev:latest
Running container...
d2c592dcee7d0f2b274b6c54be12a1d760ac68f2e9ecd0118d894fcbc39640e9
Deployment SUCCESS
Finished: SUCCESS
```

f



f

React App CI/CD Deployment using Jenkins, Docker & AWS

Project Overview

This project demonstrates end-to-end DevOps implementation:

- Dockerized React application
- CI/CD pipeline using Jenkins
- Deployment on AWS EC2

- Monitoring using Uptime Kuma
-

Tech Stack

- React
 - Docker
 - Jenkins
 - AWS EC2
 - Docker Hub
 - Uptime Kuma
-

CI/CD Flow

1. Code pushed to GitHub (dev branch)
 2. Jenkins triggers build
 3. Docker image is built
 4. Image pushed to Docker Hub (dev repo)
 5. Application deployed on EC2
 6. Monitoring enabled
-

Docker Hub Repositories

- Dev: <https://hub.docker.com/r/30727/dev>
 - Prod: <https://hub.docker.com/r/30727/prod>
-

Application URL

<http://15.206.157.134>

Monitoring

<http://:3001>

Scripts Used

build.sh

- Builds Docker image
- Tags image

- Pushes to Docker Hub

deploy.sh

- Stops old container
 - Pulls latest image
 - Runs application on port 80
-

Screenshots Included

- Jenkins Dashboard & Jobs
 - EC2 Instance & Security Group
 - Docker Hub Repositories
 - Running Application
 - Monitoring Dashboard
-

Outcome

Successfully implemented CI/CD pipeline with automated deployment and monitoring.

GitHub Repo:

<https://github.com/billa30727/devops-react-app>

App URL:

<http://15.206.157.134>

Docker Images:

30727/dev

30727/prod